

Physics 7501 (Autumn 2026)
Quantum Mechanics 1 (graduate core course)

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Abstract. In this elementary course of quantum mechanics, we will learn the mathematical basis of quantum mechanics (i.e. linear algebra), the postulates of quantum theory, and how to use quantum mechanics to understand microscopic quantum phenomena.

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0 Course Information

Instructor

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Contact Information

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Office Hours

Friday 10:00–11:00 am in PRB 2015

Lecture Logistics

- **Location:** Scott Lab N050
- **Time:** Wed. Fri. 11:10 am – 12:30 pm
- **Schedule:** See Google Spreadsheet

Teaching Assistant & Guided Group Work

- **Graduate Teaching Assistant (GTA):** Mr. Hyungjun Jeon (jeon.258@osu.edu)
- **GTA Office Hours:** Thursday 3:00–4:00 pm in PRB 2015
- **Guided Group Work (GGW) Sessions:** Thursdays 9:30 am – 11:00 am in PRB 0118. For questions regarding GGW sessions, please contact Dr. Beatriz E Burrola Gabilondo (burrolagabilondo.1@osu.edu).

Prerequisites

The course will assume physics and mathematics knowledge at the advanced undergraduate level (e.g., basic quantum mechanics, linear algebra, and differential equations), which we will review as needed.

Lecture Notes

Lecture notes will be uploaded to Carmen as PDF files generated from LaTeX sources at least 4 days before each lecture.

Main Textbooks (Recommended but Not Required)

- *Modern Quantum Mechanics* by J.J. Sakurai and Jim Napolitano (3rd Edition, Cambridge University Press, 2020)
- *Quantum Mechanics* (Vol. 3, Lectures on Theoretical Physics) by David Tong (Cambridge University Press, 2025). Problem sets available at <https://www.damtp.cam.ac.uk/user/tong/books/quantum>
- *Principles of Quantum Mechanics* by Ramamurti Shankar (2nd Edition, Plenum Press, 2011)
- *Quantum Computation and Quantum Information* by Michael Nielsen and Isaac Chuang (Cambridge University Press, 2011)

Other References

- *Lectures on Quantum Mechanics and on Topics in Quantum Mechanics*, David Tong, University of Cambridge
- *Quantum Mechanics: Non-Relativistic Theory* by Lev Landau and L. Lifshitz (2nd Edition, Pergamon Press)
- *Lectures on Quantum Mechanics* by Steven Weinberg (Cambridge University Press, 2013)
- *Quantum theory: a foundational approach* by Charis Anastopoulos (Cambridge University Press, 2024)
- *Quantum Mechanics* by Arjun Berera and Luigi Del Debbio (Cambridge University Press, 2021)

Reading Assignments

For each lecture, you will be expected to complete reading assignments before class. These will be announced on Carmen. On days when reading assignments are discussed in class, you must answer questions about the reading using Carmen "Quizzes" by 10:00 am the day of class. Your answers will not be graded for correctness; full credit is given for submitting on time. Late submissions will not be counted.

Homework

Homework will be assigned weekly on Carmen (except during holidays or exams). Submissions are due on Carmen by 11:59 pm on Fridays (or occasionally Wednesdays).

HW Quizzes

To verify individual effort in solving homework problems, a 10-minute quiz will take place during the last 10 minutes (12:20–12:30 pm) of lecture on homework due dates. Handwritten answers will be scanned and uploaded to Carmen/Gradescope.

Grading Scheme

All homework, quizzes, and exams will be graded by the GTA. Refer to Table 1 for the grading allocation.

Table 1: Grading basis for the course.

Item	Time	Assignments	Percentage
Reading Quizzes	10:00 am on lecture day	23	6%
Homework	Friday at 11:59 pm	11	18%
HW Quizzes	Last 10 mins of Friday lecture	11	18%
Mid Term Exam	Oct. 9 (Fri.), 11:30 am–1:30 pm	1	29%
Final Exam	Dec. 17 (Thu.), 12:00–2:00 pm	1	29%

Exams

- **Mid Term Exam:** In-person exam scheduled for 11:30 am–1:30 pm on Friday, Oct. 9 in PRB 1080 (Smith Seminar Room).
- **Final Exam:** In-person exam scheduled for 12:00–2:00 pm on Thursday, Dec. 17 in PRB 1080 (Smith Seminar Room).

Course Topics

- Linear algebra: mathematics of quantum mechanics
- Postulates of quantum mechanics

- Quantum entanglement
- Schrödinger equation
- Heisenberg's uncertainty principle
- One-dimensional potentials
- Introduction to scattering theory
- The harmonic oscillator and coherent states
- Squeezed states and quantum optics
- Adiabatic evolution and Berry's phase
- Quantum particles in electromagnetic fields

Shared Values

Please note The Ohio State University Shared Values.

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Academic integrity is essential to maintaining an environment that fosters excellence in teaching, research, and other educational activities. The Ohio State University and the Committee on Academic Misconduct (COAM, <https://oaa.osu.edu/resources/policies-and-procedures/committee-academic-misconduct>) expect that all students understand the University's Code of Student Conduct (<https://trustees.osu.edu/bylaws-and-rules/code>). Any activity that compromises academic integrity or subverts the educational process will be reported to COAM.

Artificial Intelligence and Academic Integrity

Generative AI tools are permitted as learning aids to support your understanding (*e.g.*, asking for alternative explanations or math reviews). However, every calculation, derivation, and explanation submitted must reflect your own understanding. You may not copy or lightly modify AI-generated solutions. AI tools are strictly prohibited during timed assessments, exams, and quizzes. When AI is permitted and used for homework, include a brief acknowledgment detailing the tool and how you verified its output.

Religious Accommodations

Ohio State accommodates students' religious beliefs and practices. Requests for accommodations must be communicated in writing to the instructor within the first 14 days of the semester. See the official policy at OAA Religious Holidays Policy.

Disability Statement & Illness

Students requesting disability accommodations should register with Student Life Disability Services (<https://slds.osu.edu/>) and notify the instructor as early as possible. If you are ill and need to miss class, notify the instructor immediately.

Intellectual Diversity

Ohio State is committed to open inquiry and intellectual diversity. Classroom discussions aim to foster critical thinking on course material without promoting any single point of view.

Grievances and Problem Solving

To resolve grievances concerning grades or academic practices, contact the instructor first, followed by the department chairperson, college dean, and provost (Faculty Rule 3335-8-23).

Creating an Environment Free from Harassment and Discrimination

The Ohio State University does not tolerate harassment, discrimination, or sexual misconduct. Reports can be submitted to the Civil Rights Compliance Office (CRCO) at <http://civilrights.osu.edu/>, by phone at 614-247-5838, or via email at civilrights@osu.edu.

1 Postulates and Mathematics of Quantum Mechanics

1.1 Key experiments that established quantum mechanics (8/26)

- Refer to Weinberg Chap 1. For further reading, try Anastopoulos Chap 2, Shankar 3.

Quantum mechanics was first motivated by a series of experiments that cannot be explained by classical mechanics and statistical physics. These exceptions to classical physics happen in the microscopic world, when the scale of physical quantities, such as energy, angular momentum and electric charge becomes comparable to certain characteristic scales.

The quantization of energy was observed in both light and atomic spectra, which motivated Planck, Einstein and others to propose a discretized (“quantized”) energy levels for photons and atoms. Bohr was able to explain the Rydberg formula[1] of hydrogen atom spectral lines under the assumption of a quantized angular momentum for electrons, which was later observed in the Stern-Gerlach experiment[2]. After de Broglie first proposed the wave nature of microscopic particles such as electrons, Davisson and Germer[3] were able to directly observe it in the diffraction pattern of electrons scattered by a nickel crystal. These experiments call for a new theory framework that defeats the intuition of classical physics.

1.1.1 Energy quantization of light in terms of photons

Young’s double-slit experiment clearly demonstrated the wave nature of light, whose energy should change continuously in classical electromagnetism. However, this point of view is seriously challenged by black-body radiation.

1.1.1.1 Black-body radiation and Planck’s formula

Consider light in a cubic box (a simplified model of a black body) of volume L^3 . A standing wave in the box satisfies the following condition for its wavevector $\vec{k} = (k_x, k_y, k_z)$:

$$\lambda_i = \frac{2\pi}{k_i} = \frac{2L}{n_i}, \quad n_i = 1, 2, 3, \dots, \quad i = x, y, z \quad (1.1)$$

Counting the number of standing wave modes in the box in terms of frequency $\nu = ck/2\pi$

$$\int dk_x dk_y dk_z = \frac{1}{8} 4\pi \int k^2 dk = \left(\frac{\pi}{L}\right)^3 \sum_{\{n_i\}} = \left(\frac{\pi}{L}\right)^3 N_{\text{modes}} \quad (1.2)$$

we have the density of states for light-wave modes per unit volume:

$$\frac{1}{L^3} \frac{dN}{d\nu} = \left(\frac{1}{\pi}\right)^3 \frac{\pi}{2} k^2 \frac{dk}{d\nu} = \frac{8\pi}{c^3} \nu^2 \quad (1.3)$$

Assuming the equipartition theorem in classical physics, i.e. each mode carries an averaged energy of $k_B T$, the black-body radiation energy per unit volume is

$$\frac{dE}{L^3 d\nu} = \frac{k_B T}{L^3} \frac{dN}{d\nu} = \frac{8\pi k_B T}{c^3} \nu^2 \quad (1.4)$$

This is known as the Rayleigh-Jeans formula and it correctly captures the long-wavelength (small ν) behavior of universal black-body radiation, first established by Kirchhoff around 1860. However, it diverges in the short wavelength limit, known as the ultraviolet catastrophe. Experimentally the Wien’s Distribution Law was observed at short wavelengths:

$$\frac{dE}{L^3 d\nu} \sim \nu^3 e^{-\alpha(T)\nu} \quad (1.5)$$

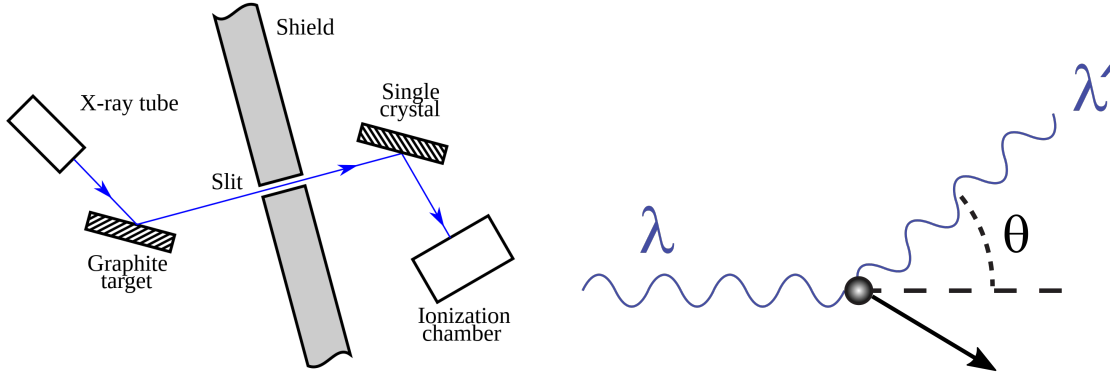


Figure 1.1: (Left) Schematic setup of Compton's experiment. (Right) The light-electron scattering process in the Compton experiment.

In order to reconcile long-wavelength Rayleigh-Jeans formula to short-wavelength Wien's law, Planck came up with his famous formula:

$$\frac{dE}{L^3 d\nu} = \frac{8\pi\nu^2}{c^3} \frac{h\nu}{e^{\alpha\nu} - 1}, \quad \alpha = \frac{h}{k_B T} \quad (1.6)$$

where the Planck constant is

$$h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s} \quad (1.7)$$

Planck's formula (1.6) indicates that the statistical-averaged energy of light at frequency ν is given by

$$U(\nu) = \frac{h\nu}{e^{h\nu/K_B T} - 1} = \frac{\sum_{n \in \mathbb{Z}_+} nh\nu e^{-nh\nu/K_B T}}{\sum_{n \in \mathbb{Z}_+} e^{-nh\nu/K_B T}} \quad (1.8)$$

under Planck's quantization hypothesis that the microscopic states of light have energy $nh\nu$ where $n \in \mathbb{Z}_+$ is a positive integer.

1.1.1.2 Einstein's "light quantum" and photoelectric effect

Motivated by Planck's quantization hypothesis, based on the assumption that light gives out energy in multiples of energy quantum $h\nu$, Einstein predicted the photoelectric effect

$$K_{\max} = eV = h\nu - \Phi \quad (1.9)$$

where $K_{\max}$ is the maximal kinetic energy of emitted photoelectrons when light of frequency ν shines on a metal, and Φ is the work function of the metal. V is minimal voltage that is required to stop photoelectron emission. This experiment was carried out by Millikan[4], who obtained Planck's constant

$$h = e \frac{dV}{d\nu} \quad (1.10)$$

1.1.1.3 Compton effect demonstrated quantization of light in terms of photons

Discovered by Arthur Compton in 1923[5], when the energy of the X ray photon (~ 17 keV) was significantly larger than the binding energy of the atomic electron, the electrons could be treated as being free after scattering. In this case, a *Compton shift* between the wavelengths of

injected light (λ) and the scattered light (λ') was observed to satisfy the following relation with the scattering angle θ (see Fig. 1.1):

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta) \quad (1.11)$$

where h is the Planck's constant.

The Compton shift can be explained by energy and momentum conservation, by assuming the photon energy of $E = h\nu = hc/\lambda$.

1.1.2 Quantization of atomic energy levels

1.1.2.1 Rydberg formula for the spectral lines in a hydrogen atom

After the discovery of dark lines in the spectrum of the sun at the beginning of 18th century, it was later realized that hot atomic gases absorb and emit light only at certain definite frequencies. Building on works of Balmer and others, Rydberg discovered the following rule[1] for the frequency of emitted light from hydrogen gases:

$$\nu = \frac{c}{\lambda} = cR_H \left(\frac{1}{n^2} - \frac{1}{m^2} \right), \quad m, n \in \mathbb{Z}_+. \quad (1.12)$$

This clearly demonstrated the discreteness of atomic energy levels $E_n \propto \frac{1}{n^2}$, further motivating Bohr's theory of the hydrogen atom.

1.1.2.2 Franck-Hertz experiment

In the Franck-Hertz experiment (1914) illustrated in Fig. 1.2, in a vacuum tube fitted with three electrodes (an electron-emitting, hot cathode; a metal mesh grid; and an anode), when a grid voltage is applied between the cathode and the anode, the electric current through the tube filled with thin mercury vapours, due to electrons that pass through the grid and reach the anode, is measured in the experiment. As shown in the middle figure, the current increases rapidly at low voltages, and then sharply drops almost back to zero at 4.9 V, with similar effects observed as 9.8 V and other multiples of 4.9 V. This can be understood by the quantized energy levels of mercury atoms. Inside the single mercury droplet in the vacuum tube, as shown in Fig. 1.2.

1.1.3 Quantization of angular momentum

1.1.3.1 Bohr's explanation of hydrogen atom spectral lines

Rydberg's rule[1] indicates that the photon energy emitted by the hydrogen atom is given by

$$h\nu = E_m - E_n, \quad E_n = -\frac{hcR_H}{n^2}, \quad n \in \mathbb{Z}_+. \quad (1.13)$$

How to explain the $1/n^2$ quantization of hydrogen atom energy levels? According to classical mechanics, for an electron in the Coulomb force field of hydrogen nucleus, its speed v and radius r satisfies

$$\frac{mv^2}{r} = \frac{ke^2}{r^2} \implies E = \frac{1}{2}mv^2 - \frac{ke^2}{r} = -\frac{ke^2}{2r} \quad (1.14)$$

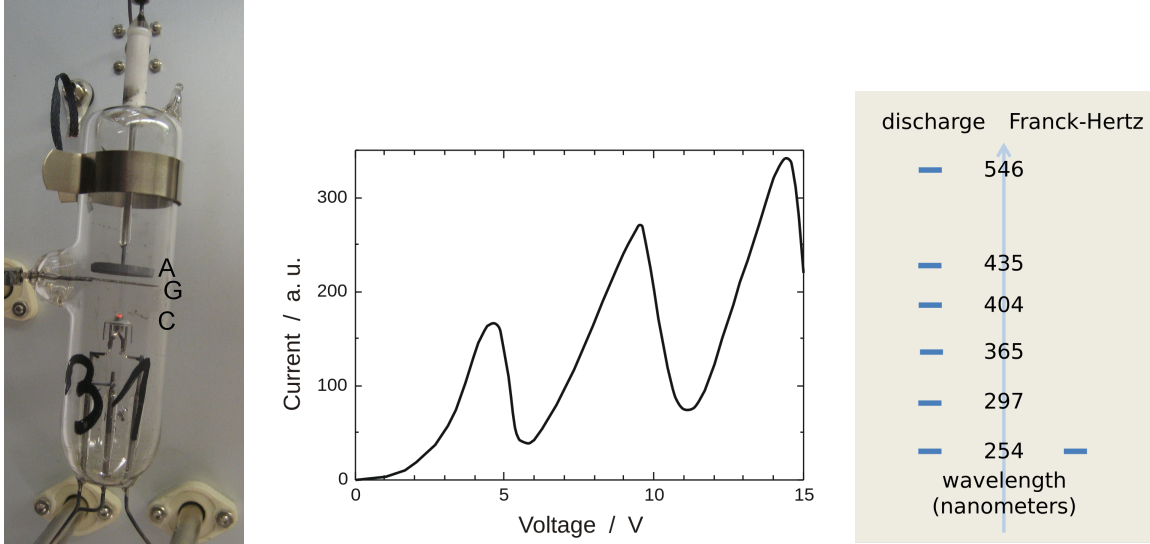


Figure 1.2: (Left) Photograph of a vacuum tube used for the Franck-Hertz experiment. (Middle) Anode current (arbitrary units) versus the grid voltage relative to the cathode. (Right) Wavelengths of light emitted by a mercury vapor discharge and by a Franck-Hertz tube in operation at 10 V. (Picture from Wikipedia)

where k is the Coulomb constant. To fit the n^2 dependence of the orbit radius r , Niels Bohr came up with the following quantization condition for angular momentum:

$$mvr = n\hbar \quad (1.15)$$

where $\hbar$ turns out to be $h/2\pi$. This quantization condition leads to the famous energy levels of electrons in a hydrogen atom:

$$E_n = -\frac{mk^2e^4}{2\hbar^2} \frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2} \quad (1.16)$$

1.1.3.2 Stern-Gerlach experiment demonstrated the quantization of angular momentum

In 1922 Stern and Gerlach performed an experiment[2] measuring the deflection of silver atoms by an inhomogeneous magnetic field, as illustrated in Fig. 1.3. Due to the interaction energy $U = -\vec{\mu} \cdot \vec{B}$ of a magnetic moment $\vec{\mu}$ under a magnetic field $\vec{B} = B_z \hat{z}$, a non-uniform magnetic field exerts a force $\vec{F} = -\vec{\nabla}U = \mu_z \vec{\nabla}B_z$ on the moment-carrying silver atoms, therefore deflecting the silver atom beams in the inhomogeneous magnetic field. As shown in Fig. 1.3, in contrast a continuous range of values of the magnetic moment as expected in classical mechanics, the experimental observations pointed to quantization of magnetic moment $\mu_z = g\mu_B S_z/\hbar$, where we defined μ_B as the Bohr magneton:

$$\mu_B = \frac{\hbar e}{2m_e} \quad (1.17)$$

$g \approx 2$ as the electron g -factor and

$$S_z = \pm \frac{\hbar}{2} \quad (1.18)$$

is the quantized spin angular momentum of a silver atom observed in Stern-Gerlach experiment.

This experiment demonstrated that in addition to energy level quantization, other physical observable such as spin angular momentum are also quantized in microscopic scales such as

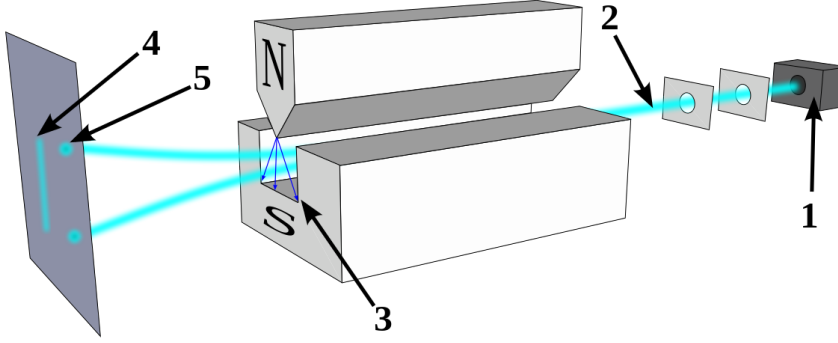


Figure 1.3: Schematic setup of Stern-Gerlach experiment: Silver atoms travelling through an inhomogeneous magnetic field, being deflected up or down depending on their spin. (1) furnace, (2) beam of silver atoms, (3) inhomogeneous magnetic field, (4) the classically expected result, (5) the observed result.

in silver atoms. Moreover, we can consider sequential Stern-Gerlach experiments shown in Fig. 1.4, where silver atoms are deflected by multiple inhomogeneous magnetic fields in a sequence. Remarkably, unlike the classical case of light passing through polarizers, which filter out everything after two polarizers with perpendicular polarization axes, two quantum mechanical “beam splitters” with perpendicular magnetic field directions still allows two splitted beams to pass through, as illustrated in the middle and bottom parts of Fig. 1.4.

1.1.4 Davisson-Germer experiment demonstrated the wave nature of electrons

In 1924 de Broglie formally proposed the wave nature of electrons and other particles, in terms of their de Broglie wavelengths:

$$\lambda = \frac{h}{p} \quad (1.19)$$

where p is the momentum of the particle. If this proposal were true, it means that microscopic particles such as electrons should exhibit interference phenomena in the same way as acoustic and light waves.

In 1927 Davisson and Germer[3] performed an experiment to observe the diffraction pattern of electrons scattered by the surface of a crystal of nickel metal, as shown in Fig. 1.5. The highest intensity was observed at an angle $\theta = 50^\circ$ with a voltage of 54 V, giving the electrons a kinetic energy of 54 eV. To be compatible with the Bragg’s law

$$n\lambda = 2d \sin(90^\circ - \theta/2), \quad n \in \mathbb{Z}_+. \quad (1.20)$$

with the lattice spacing of nickel $d = 0.091$ nm, the $n = 1$ Bragg peak in (1.20) exactly corresponds to the de Broglie wavelength (1.19) of electrons. We will examine the experimental

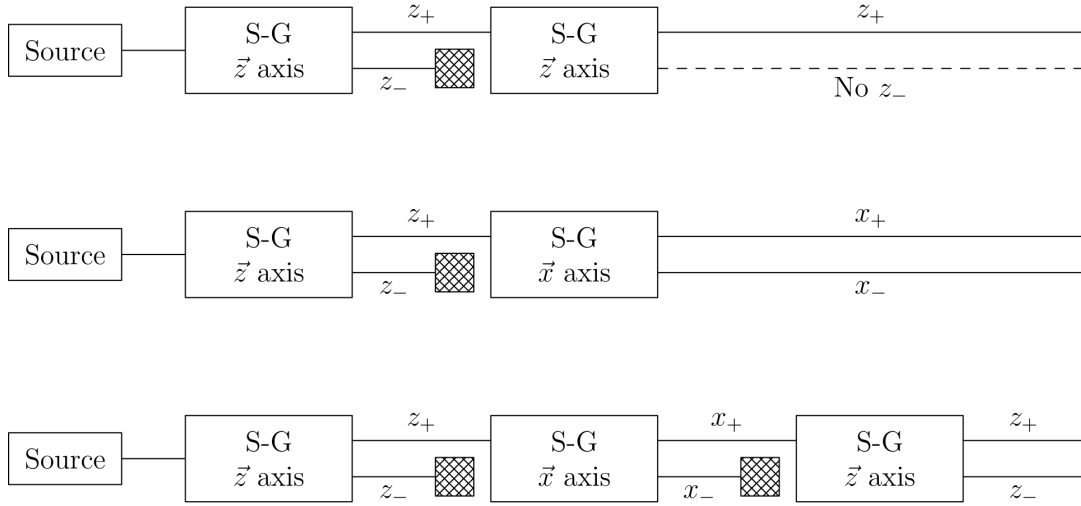


Figure 1.4: Sequential Stern-Gerlach experiments. (Picture from S&N)

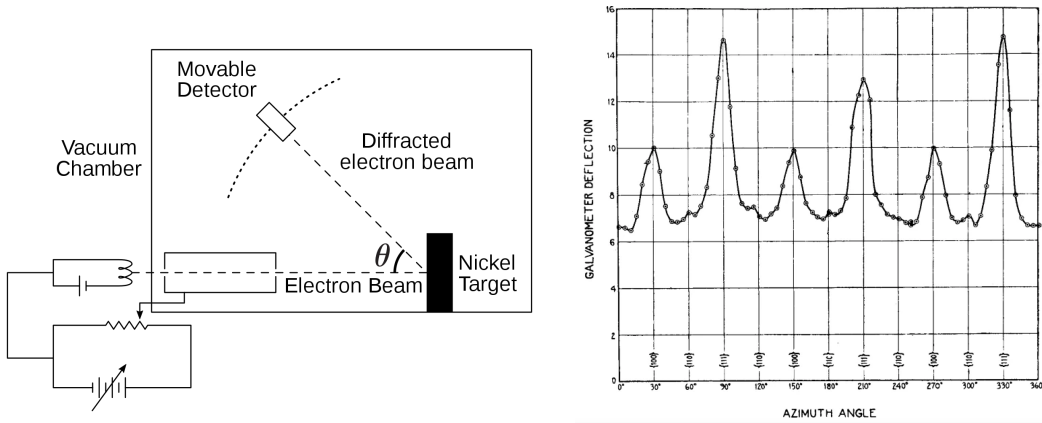


Figure 1.5: (Left) Schematic setup of Davisson-Germer experiment. (Right) Intensity of electron scattering vs. azimuth angle at 54 volts, co-latitude $\theta = 50^\circ$.

1.1 Key experiments that established quantum mechanics (8/26)

results in Fig. 1.5 more carefully in Homework 1.

1.2 Hilbert space and state vectors (8/28)

- Refer to N&C 2.1, Tong 3.1, Shankar 1.1-1.4

How to explain the above experimental observations? 4 postulates, derived (or more precisely, guessed) after a long process of trial and error, form the foundation for a theory framework that explains all observed experimental phenomena, known as quantum mechanics. Below we will introduce these postulates one by one, along the way of introducing linear algebra as the mathematics of quantum mechanics.

Postulate 1.1. *Associated to any isolated physical system is a complex vector space with inner product (i.e. a Hilbert space) known as the state space of the system. The system is completely described by its state vector, which is a unit vector in the system's state space.*

The key mathematical framework for quantum mechanics is linear algebra. The basic objects of linear algebra are vector spaces:

A *vector space* over a *scalar field* $\mathbb{F}$ is a set $\mathbb{V}$ of objects (called *vectors*) equipped two binary operations: a *vector addition* (“+”)

$$|v\rangle + |w\rangle \in \mathbb{V}, \quad \forall |v\rangle, |w\rangle \in \mathbb{V}. \quad (1.21)$$

and a *scalar multiplication*:

$$a|v\rangle \in \mathbb{V}, \quad \forall |v\rangle \in \mathbb{V}, a \in \mathbb{F}. \quad (1.22)$$

Mathematically, the set $\mathbb{V}$ forms an abelian group under vector addition operation and the scalar multiplication defines a ring homomorphism from the field $\mathbb{F}$ into the endomorphism ring of this abelian group. We will break down these terminology below.

It is worthwhile to define a group since we will study symmetry in quantum mechanics later. A group is a set $V = \{u, v, \dots\}$ equipped with a *closed* binary operation $+$: $V \times V \rightarrow V$

$$u + v = w \in V, \quad \forall u, v \in V. \quad (1.23)$$

where 3 group axioms are satisfied:

(1) *Identity element*:

$$\exists 0 \in V, \text{ s.t. } \forall u \in V, u + 0 = 0 + u = u. \quad (1.24)$$

(2) *Inverse*:

$$\forall u \in V, \exists (-u) \in V, \text{ s.t. } u + (-u) = 0. \quad (1.25)$$

(3) *Associativity*:

$$(u + v) + w = u + (v + w), \quad \forall u, v, w \in V. \quad (1.26)$$

An Abelian group is a group whose addition operation is commutative:

$$u + v = v + u, \quad \forall u, v \in V. \quad (1.27)$$

The scalar multiplication is another map $\mathbb{F} \times \mathbb{V} \rightarrow \mathbb{V}$ that is *associative*

$$a(b|v\rangle) = (ab)|v\rangle, \quad \forall a, b \in \mathbb{F}, |v\rangle \in \mathbb{V} \quad (1.28)$$

and *distributive in both the vectors and the scalars*:

$$a(|u\rangle + |v\rangle) = a|u\rangle + a|v\rangle, \quad (1.29)$$

$$(a + b)|v\rangle = a|v\rangle + b|v\rangle, \quad \forall a, b \in \mathbb{F}, |u\rangle, |v\rangle \in \mathbb{F}. \quad (1.30)$$

The above conditions satisfied by vector addition and scalar multiplication are summarized well in Chapter 1.1 of Shankar.

Of particular interests to quantum mechanics is a *Hilbert space*, which is a (n -dimensional) vector space $\mathbb{C}^n$ over the field $\mathbb{C}$ of complex numbers, equipped with a map $\mathbb{V} \times \mathbb{V} \rightarrow \mathbb{C}$ known as an *inner product*:

$$\langle u|v\rangle \in \mathbb{C}, \quad \forall |u\rangle, |v\rangle \in \mathbb{V}. \quad (1.31)$$

In the Dirac notation, $\langle u|$ is known as a *bra*, and $|v\rangle$ is known as a *ket*. The inner product is *skew-symmetric*

$$\langle u|v\rangle = \langle v|u\rangle^* \quad (1.32)$$

positive semidefinite

$$\langle v|v\rangle \geq 0, \quad \langle v|v\rangle = 0 \iff |v\rangle = 0. \quad (1.33)$$

and *linear in ket*

$$\langle u|(\sum_i \lambda_i |v_i\rangle) = \sum_i \lambda_i \langle u|v_i\rangle \quad (1.34)$$

These properties are summarized in Chapter 1.2 of Shankar and Chapter 2.1.4 of N&C.

Note that an inner product also defines a norm for each vector in vector space $\mathbb{V}$:

$$||v\rangle| \equiv \sqrt{\langle v|v\rangle} \quad (1.35)$$

Two important properties of an inner product are

Theorem 1.2. *Cauchy-Schwarz inequality*:

$$\langle v|v\rangle \langle u|u\rangle \geq |\langle u|v\rangle|^2 \quad (1.36)$$

The equality holds if and only if $|v\rangle$ and $|w\rangle$ are linearly dependent.

Theorem 1.3. *Triangle inequality for the norms of vectors*:

$$||v\rangle + |w\rangle| \leq ||v\rangle| + ||w\rangle| \quad (1.37)$$

Triangle inequality can be derived from the Cauchy-Schwarz inequality, which is a part of Homework 2.

Strictly speaking, in addition to being a vector space equipped with an inner product (known as an inner product space), a Hilbert space must also be *complete* in the sense that every Cauchy sequence $\{v^{(k)}\}$ of vectors in $\mathbb{V}$ has a limit that also lies in $\mathbb{V}$:

$$\lim_{k \rightarrow \infty} v^{(k)} \in \mathbb{V}, \quad \text{if } \forall \epsilon > 0, \quad \exists N \in \mathbb{Z}_+, \text{ s.t. } |v^{(m)} - v^{(n)}| < \epsilon, \quad \forall m, n > N. \quad (1.38)$$

For an inner product space of finite dimensions on any complete field $\mathbb{F}$, such as real and complex numbers $\mathbb{F} = \mathbb{R}, \mathbb{C}$, it is automatically a Hilbert space. Below are a few examples of Hilbert spaces, and inner product spaces that are not Hilbert spaces:

1 Postulates and Mathematics of Quantum Mechanics

- n -dimensional Hilbert space on complex numbers: $\mathbb{V} \simeq \mathbb{C}^n$
- $n \times n$ complex matrices equipped with the Hilbert-Schmidt inner product forms a Hilbert space $L_{\mathbb{C}^n}$ (see Homework 1)
- n -dimensional Hilbert space on real numbers: $\mathbb{V} \simeq \mathbb{R}^n$
- n -dimensional vector space $\mathbb{V} \simeq \mathbb{Q}^n$ on rational numbers is not a Hilbert space because it is not complete, since a Cauchy sequence of rational numbers can converge to an irrational number.
- The inner product space of continuous square-integrable complex functions on interval $[a, b]$ with the following L^2 inner product:

$$\langle f_1 | f_2 \rangle \equiv \int_a^b f_1^*(x) f_2(x) dx \quad (1.39)$$

is not a Hilbert space due to its incompleteness: a Cauchy sequence of continuous functions can be discontinuous. Its completion is the Hilbert space $L^2[a, b]$.

- For the same reason of incompleteness, the inner product space of polynomials on $[a, b]$ with the same L^2 inner product is not a Hilbert space, even though it's a subspace of $L^2[a, b]$.

A Hilbert space forms the *state space* described in Postulate 1.1. With the help of the inner product structure, we can always choose a complete set of orthonormal basis $\{|i\rangle\}$ that expands the full state space $\mathbb{V}$, such that

$$|v\rangle = \sum_i v_i |i\rangle, \quad \forall |v\rangle \in \mathbb{V}; \quad \langle i | j \rangle = \delta_{i,j}, \quad v_i = \langle i | v \rangle. \quad (1.40)$$

where $\delta_{i,j}$ is the Kronecker delta function. Given a n -dimensional vector space V , one can always find a set of n linearly independent vectors $\{e_j | 1 \leq j \leq n\}$, known as a basis of vector space V , such that any vector $v \in V$ can be uniquely expressed as a linear superposition of the basis vectors: $v = \sum_j v_j e_j$ with $v_j \in \mathbb{F}$. Given the inner product structure in a Hilbert space, a set of complete orthonormal basis can be constructed following the Gram-Schmidt procedure, discussed in details in Chapter 1.2 of Shankar and Chapter 2.1.4 of N&C. In such a complete orthonormal basis of Hilbert space $\mathbb{V} = \mathbb{C}^n$, a Dirac ket $|v\rangle$ can be represented by a column vector $\vec{v} = (v_1, \dots, v_n)^T$, a Dirac bra represented by a row vector, and the inner product reduces to the familiar form in $\mathbb{C}^n$:

$$\langle u | v \rangle = \sum_i u_i^* v_i = (u_1^*, \dots, u_n^*) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \quad (1.41)$$

The column vector $(v_1, \dots, v_n)$ is known as the *wavefunction* of state $|v\rangle$ in the basis set $\{|i\rangle\}$.

A special set of complete orthonormal basis that plays a particularly important role in quantum mechanics is the eigenstates $|x\rangle$ of position operator $\hat{x}$, where we consider one dimension for simplicity. For any normalized state $|\psi\rangle$, its continuous-variable wavefunction

$$\psi(x) \equiv \langle x | \psi \rangle, \quad \int dx |\psi(x)|^2 = \langle \psi | \psi \rangle = 1. \quad (1.42)$$

determines the probability of detecting the quantum-mechanical particle at position x :

$$\text{Pr}(x) = |\psi(x)|^2, \quad \int dx \text{Pr}(x) = 1. \quad (1.43)$$

1.3 Linear operators and observables (9/2)

- Refer to N&C 2.1; Tong 3.2, 4.1; S&N 1.3-1.4; Shankar 1.5-1.6

Given two vector spaces $\mathbb{V}$ and $\mathbb{W}$, we can also define a linear operator $\hat{O}$ as any function that maps the vector space $\mathbb{V}$ to $\mathbb{W}$, which is linear in its inputs:

$$\hat{O}(\sum_i a_i |v_i\rangle) = \sum_i a_i \hat{O}(|v_i\rangle) \quad (1.44)$$

In the case of $\mathbb{V} = \mathbb{W}$ we call it a linear operator $\hat{O} \in \mathbb{L}_{\mathbb{V}}$ on vector space $\mathbb{V}$, which is of particular interest to quantum mechanics.

Using a complete set of orthonormal basis $\{|j\rangle | 1 \leq j \leq \dim \mathbb{V}\}$ for Hilbert space $\mathbb{V}$, each linear operator has a matrix representation:

$$\hat{O} = \sum_{a,b} O_{a,b} |a\rangle \langle b|, \quad O_{a,b} \equiv \langle a | \hat{O} | b \rangle \quad (1.45)$$

In particular, for the identity matrix which maps every vector to itself, we have the **completeness relation** for basis $\{|i\rangle\}$, also known as the **resolution of identity**:

$$\sum_i |i\rangle \langle i| = \hat{1}. \quad (1.46)$$

In the case of continuous variables, e.g. coordinates for a particle in free space, the summation will be replaced by integral:

$$\int dx |x\rangle \langle x| = \hat{1} \quad (1.47)$$

where $|x\rangle$ is an eigenstate of position operator $|x\rangle$ with eigenvalue x .

We can also define the product of two linear operators $\hat{M}, \hat{N} \in \mathbb{L}_{\mathbb{V}}$:

$$\hat{M}\hat{N} = \sum_{a,b,c} M_{a,b} N_{b,c} |a\rangle \langle c| \quad (1.48)$$

their commutator

$$[\hat{M}, \hat{N}] = \hat{M}\hat{N} - \hat{N}\hat{M} \quad (1.49)$$

and anti-commutator

$$\{\hat{M}, \hat{N}\} = \hat{M}\hat{N} + \hat{N}\hat{M} \quad (1.50)$$

and the *Hermitian conjugate* (or *Hermitian adjoint*) of an operator

$$\hat{O}^\dagger = \sum_{a,b} O_{b,a}^* |a\rangle \langle b| = (O^T)^* = (O^*)^T \quad (1.51)$$

Specifically in Postulate 1.9 about quantum (projective) measurements, each physical observable corresponds to a Hermitian linear operator, represented as a Hermitian matrix in the orthonormal basis:

$$\hat{M} = \hat{M}^\dagger = \sum_{a,b} M_{a,b} |a\rangle \langle b|, \quad (M^\dagger)_{a,b} = M_{b,a}^* = M_{a,b} \equiv \langle a | \hat{M} | b \rangle. \quad (1.52)$$

The diagonal form

$$\hat{O} = \sum_a O_a |a\rangle\langle a| \quad (1.53)$$

where $\{|a\rangle\}$ is a complete set of orthonormal basis of the Hilbert space, is also known as the *spectral decomposition* of a matrix $\hat{O}$. If there is a complete set of orthonormal basis $\{|a\rangle\}$ such that the linear operator $\hat{O}$ can be rewritten as the diagonal form (1.53), the matrix $\hat{O}$ is called “diagonalizable”.

Not all matrices are diagonalizable (i.e. have a diagonal form), two non-diagonalizable examples are

$$\hat{M}_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \hat{M}_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad (1.54)$$

$\hat{M}_1$ is known as a Jordan block with only 1 eigenvector, while $\hat{M}_2$ has two eigenvectors with different eigenvalues, but not orthogonal to each other. Generally, any complex square matrix can be decomposed into tensor sums of Jordan blocks by similarity transformations.

Below are a few useful theorems on the diagonalization of matrices:

Theorem 1.4. The spectral decomposition theorem: A linear operator $\hat{O}$ on vector space $\mathbb{V}$ is diagonal w.r.t. some orthonormal basis of $\mathbb{V}$ if and only if operator $\hat{O}$ is normal i.e. $\hat{O}^\dagger \hat{O} = \hat{O} \hat{O}^\dagger$.

Refer to N&C Box 2.2 for proof of this theorem.

Theorem 1.5. A normal matrix is Hermitian if and only if it has real eigenvalues.

Theorem 1.6. The eigenkets corresponding to different eigenvalues of a Hermitian matrix are orthogonal.

Theorem 1.7. A positive linear operator must be Hermitian.

Theorem 1.8. Simultaneous diagonalization theorem: Two Hermitian linear operators $\hat{A}$ and $\hat{B}$ can be diagonalized simultaneously if and only if $[\hat{A}, \hat{B}] = 0$.

In other words, there is a complete set of simultaneous eigenket $\{|a, b\rangle\}$ such that

$$\hat{A}|a, b\rangle = a|a, b\rangle, \quad \hat{B}|a, b\rangle = b|a, b\rangle. \quad (1.55)$$

if and only if $[\hat{A}, \hat{B}] = 0$.

An Hermitian operator can be diagonalized into a special orthonormal basis $\{|i\rangle\}$, known as its eigenbasis, in which the linear operator is represented by a diagonal matrix

$$\hat{M} = \sum_i \lambda_i |i\rangle\langle i| \quad (1.56)$$

If we use the set $\{m\}$ to label all different (real) eigenvalues of matrix $\hat{M}$, we can rewrite the above diagonal form (1.56) as

$$\hat{M} = \sum_m m \hat{P}_m, \quad \hat{P}_m = \sum_j \delta_{\lambda_j, m} |j\rangle\langle j| \quad (1.57)$$

where operators $\hat{P}_m$ are known as *projection operators* (or *projectors*) onto the eigenspace of $\hat{M}$ with eigenvalue m . Mathematically, a projector $\hat{P}$ is defined as a linear operator satisfying:

$$\hat{P}^2 = \hat{P} \quad (1.58)$$

You will learn more about projectors in HW2.

Given a state $|\psi\rangle$ and a physical observable $\hat{M}$, the (probability-weighted) *expectation value* $\langle\hat{M}\rangle$ of operator $\hat{M}$ w.r.t. state vector $|\psi\rangle$ is

$$\langle\hat{M}\rangle \equiv \frac{\langle\psi|\hat{M}|\psi\rangle}{\langle\psi|\psi\rangle} \quad (1.59)$$

Naturally, the expectation value of any physical observable is a real number, since any physical observable corresponds to a Hermitian operator.

We note that the expectation value (1.59) of any observable is the same for two linearly dependent states $|\psi\rangle$ and $\alpha|\psi\rangle$ for any complex scalar $\alpha \in \mathbb{C}$. This means $|\psi\rangle$ and $\alpha|\psi\rangle$ describe the same physical state. In other words, in a N -dimensional Hilbert space $\mathbb{V} = \mathbb{C}^N$, for a generic state

$$|\psi\rangle \equiv \sum_{j=1}^N \psi_j |j\rangle \quad (1.60)$$

where $\{|j\rangle | 1 \leq j \leq N\}$ is a complete set of orthonormal basis of $\mathbb{V}$, although $\{\psi_j \in \mathbb{C} | 1 \leq j \leq N\}$ correspond to $2N$ independent real variables, only $2N - 2$ of them are physically measurable, because the global phase factor and normalization leads to 2 redundancies that cannot be physically observed.

In the simplest example of a single qubit, we have $\mathbb{V} = \mathbb{C}^2$ for the $N = 2$ -dimensional Hilbert space. There are 4 linearly independent Hermitian operators in $\mathbb{C}^2$, whose matrix representations are

$$\hat{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.61)$$

As you will learn in HW1, they (up to normalization) form a complete set of orthonormal basis for the Hilbert space $L_{\mathbb{C}^2}$ of linear operators. In the matrix representation, a generic state vector can be written as

$$|\psi\rangle = \alpha \begin{pmatrix} \cos \theta e^{i\phi} \\ \sin \theta \end{pmatrix}, \quad \alpha \in \mathbb{C} \quad (1.62)$$

θ and ϕ are the two variables in the state vector that can be physically measured. One can carry out measurements for two observables to extract θ and ϕ :

$$\langle\hat{\sigma}_z\rangle = \cos(2\theta), \quad \langle\hat{\sigma}_x\rangle = \sin(2\theta) \cos \phi \quad (1.63)$$

1.3.1 Appendix: Functions of operators/matrices

A function $f(\hat{M})$ of a square matrix $\hat{M}$ can be defined by the following Taylor expansion

$$f(\hat{M}) = \sum_{n=0}^{\infty} \frac{\hat{M}^n}{n!} \frac{d^n f(x)}{dx^n} \Big|_{x=0} \quad (1.64)$$

if the Taylor expansion converges at $x = 0$. In quantum mechanics we will mostly encounter square matrices that can be diagonalized, such as Hermitian and unitary matrices.

For a diagonalizable matrix $\hat{M}$ with eigenkets $\{|j\rangle\}$ and eigenvalues $\{\lambda_j\}$, the matrix can be written as

$$\hat{M} = \sum_j \lambda_j |j\rangle\langle j| \quad (1.65)$$

and a function of this diagonalizable matrix can be defined as

$$f(\hat{M}) = \sum_j f(\lambda_j) |j\rangle\langle j| \quad (1.66)$$

1.4 Projective measurements (9/4)

- Refer to N&C 2.2.4-2.2.7, Tong 3.4, 4.2, 16.4.1

Postulate 1.9. A *projective measurements* is described by a physical observable, $\hat{M} = \hat{M}^\dagger$ on the Hilbert space of the system being observed. The observable has a spectral decomposition

$$\hat{M} = \sum_m m \hat{P}_m \quad (1.67)$$

where $\hat{P}_m$ is the projector onto the eigenspace of M with eigenvalue m . The possible outcomes of the measurement correspond to the eigenvalues m of the observable. Upon measuring state $|\psi\rangle$, the probability of getting result m is given by

$$\text{Pr}(m) = \langle \psi | \hat{P}_m | \psi \rangle \quad (1.68)$$

If the outcome m occurred, the state of the quantum system collapsed into a new state

$$\hat{P}_m |\psi\rangle / \sqrt{\text{Pr}(m)} \quad (1.69)$$

immediately after the measurement.

Postulate 1.9 is in fact only a special form of quantum measurements, which will be discussed in detail in later parts of this course. Most generally, a generalized quantum measurement is described as follows:

Postulate 2a: A **generalized quantum measurement** is described by a collection $\{M_m\}$ of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the (normalized) state of the quantum system is $|\psi\rangle$ immediately before the measurements, then the probability that result m occurs is given by

$$\text{Pr}(m) = \langle \psi | M_m^\dagger M_m | \psi \rangle \quad (1.70)$$

and the state of the system after the measurement is $\frac{M_m |\psi\rangle}{\sqrt{\langle \psi | M_m^\dagger M_m | \psi \rangle}}$. The measurement operators satisfy the completeness equation

$$\sum_m M_m^\dagger M_m = \hat{1}. \quad (1.71)$$

When we choose $\hat{M}_m$ to be projector $\hat{P}_m$ into the eigenspace of observable $\hat{M}$ with eigenvalue m , the above generalized measurement reduces to the a projective measurement. Now it also becomes clear that the expectation value of observable $\hat{M}$ w.r.t. to (normalized) state vector $|\psi\rangle$ is nothing but the probability-weighted average of measurement outcomes:

$$\langle \hat{M} \rangle \equiv \langle \psi | \hat{M} | \psi \rangle = \sum_m m \times \text{Pr}(m) \quad (1.72)$$

One can also find out the *standard deviation* of measuring observable $\hat{M}$ in state $|\psi\rangle$:

$$\Delta(M) \equiv \sqrt{\langle (\hat{M} - \langle \hat{M} \rangle)^2 \rangle} = \sqrt{\langle \hat{M}^2 \rangle - \langle \hat{M} \rangle^2} \quad (1.73)$$

A significant consequence of the collapse part in Postulate 1.9 is the following: starting from any state vector $|\psi\rangle$, if we measure the same observable $\hat{M}$ multiple times, the measurement

outcome does not change any more after the 1st measurement. This is because the state already collapse into an eigenstate of $\hat{M}$ with some eigenvalue m after the 1st measurement.

Another interesting aspect comes from *incompatible measurements* of physical observables that do not commute with each other. If we perform two measurements $\hat{A}$ and $\hat{B}$ that satisfy $[\hat{A}, \hat{B}] \neq 0$, for a generic quantum state, the measurement outcome depend on the order of the two measurements: i.e. measuring $\hat{A}$ after $\hat{B}$ versus $\hat{B}$ after $\hat{A}$ will yield different results. For example, starting from the eigenstate $|\uparrow\rangle$ of Pauli matrix σ_z , if we measure σ_z and then σ_x , there is a probability of 1/2 for us to end up with +1 outcomes for both measurements. However, if σ_x is measured before the σ_z measurement, there is only a probability of 1/4 of +1 outcomes for both measurements.

1.5 Unitary evolution of a closed quantum system (9/9)

- Refer to S&N 2.1; Tong 3.3; N&C 2.1.8, 2.2.2, 4.2; Shankar 4.3

In addition to Postulates 1.1 and 1.9 introduced earlier, we have two more postulates, one regarding how a quantum system evolves with time:

Postulate 1.10. *The time evolution of the state of a closed quantum system is described by the Schrodinger equation:*

$$i\hbar \frac{d|\psi\rangle}{dt} = \hat{H}|\psi\rangle \quad (1.74)$$

where $\hbar = h/2\pi \approx 1.05 \times 10^{-34}$ J·s is a constant, and $\hat{H}$ is a fixed Hermitian operator known as the Hamiltonian of the closed system.

Postulate 1.10 states that the state vector associated with a closed quantum system evolves following the Schrodinger equation (1.74). This postulate can be states in an equivalent manner:

Postulate 3a *The evolution of a closed quantum system is described by a unitary transformation. That is, the state $|\psi(t_2)\rangle$ of the system at time t_2 is related to the state $|\psi(t_1)\rangle$ at time t_1 by a unitary operator U which only depends on the times t_1 and t_2 :*

$$|\psi(t_2)\rangle = U(t_1, t_2)|\psi(t_1)\rangle, \quad U(t_1, t_2) = e^{-i(t_2-t_1)\hat{H}/\hbar} \quad (1.75)$$

One can further generalize this to a time-dependent Hamiltonian $\hat{H}(t)$ where we have

$$U(t_1, t_2) = \mathcal{T}e^{-i \int_{t_1}^{t_2} \hat{H}(t) dt / \hbar} \quad (1.76)$$

where $\mathcal{T}$ denotes time ordering in the integral, as explained in S&N 2.1.

In linear algebra, a unitary operator $\hat{U}$ is defined as any linear operator satisfying

$$\hat{U}^\dagger \hat{U} = \hat{1}. \quad (1.77)$$

which also implies $\hat{U}\hat{U}^\dagger = \hat{1}$. Due to the spectrum decomposition theorem, a unitary operator can be diagonalized, and its eigenvalues all have modulus 1:

$$\hat{U}|\psi\rangle = e^{i\theta}|\psi\rangle \quad (1.78)$$

Three examples of unitary operators in quantum mechanics are the following: (1) unitary time evolution operator $U(t_1, t_2) = e^{-i(t_2-t_1)\hat{H}/\hbar}$ for a closed quantum system discussed above; (2) a symmetry operator in a quantum system as we will discuss later in this course; (3) a quantum (logic) gate in quantum computing. For instance, in a two-dimensional closed quantum system, also known as a qubit, the common quantum gates are summarized in Table 1.1.

Starting from a complete set of orthonormal basis vectors $\{|i\rangle\}$ of Hilbert space $\mathbb{V}$, after the action of a unitary operator $\hat{U}$, we can obtain another complete set of orthonormal basis vectors $\{|i'\rangle = \hat{U}|i\rangle\}$ of the same Hilbert space. In other words, each unitary operator implements a change of basis in the Hilbert space. Therefore, the unitary time evolution merely performs a rotation for vectors in the Hilbert space, since it only changes the state itself but not the inner product of any two states.

In the case of unitary time evolution $U(t_1, t_2)$, the Hamiltonian $\hat{H}$ of a closed quantum system plays a crucial role. In its spectral decomposition:

$$\hat{H} = \sum_a E_a |a\rangle\langle a| \quad (1.79)$$

Single-qubit gate	Symbol	Unitary matrix
Pauli-X (bit flip)	X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Pauli-Y	Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Pauli-Z (phase flip)	Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Hadamard	H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
Phase	S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$
$\pi/8$	T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$

Table 1.1: Common single-qubit gates and associated unitary matrices.

the eigenkets $|a\rangle$ are called the *energy eigenstates*, or *stationary states*, and the eigenvalue E_a is called the energy of state $|a\rangle$. In particular, the eigenstate with the lowest energy is known as the *ground state*, while other states with higher energy are known as *excited states*. According to Schrodinger equation (1.74), a generic state vector

$$|\psi\rangle = \sum_a c_a |a\rangle \quad (1.80)$$

evolves as

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi\rangle = \sum_a c_a e^{-iE_a t/\hbar} |a\rangle \quad (1.81)$$

and the different oscillation frequency $\omega_a = E_a/\hbar$ in different components of state $|\psi(t)\rangle$ can result in oscillations of physical observables. Two such examples, i.e. (1) spin precession; and (2) neutrino oscillation are discussed in detail in S&N 2.1, and will be a part of HW3. In particular, a physical observable you will encounter a lot in the future is the *transition probability*:

$$P(i \rightarrow f) \equiv |\langle f | e^{-i\hat{H}t/\hbar} | i \rangle|^2 \quad (1.82)$$

which measures the probability that an initial state $|i\rangle$ evolves into a final state $|f\rangle$ after time t . This quantity is directly related to the outgoing beam intensity in scattering experiments.

The transition probability is the modulus square of the *propagator*

$$G(i, f : t) \equiv \langle f | e^{-i\hat{H}t/\hbar} | i \rangle \quad (1.83)$$

which plays very important conceptual and computational roles in quantum mechanics. We will learn more about the propagator in HW3.

1.5.1 Schrodinger vs. Heisenberg pictures

Regarding the time dependence of physical observables, such as the expectation value of a Hermitian operator $\hat{O}$ at time t , there are two equivalent ways to approach this problem.

In the Schrodinger picture, where the state vector $|\psi\rangle$ evolves with time while the operator $\hat{O}$ remains a constant:

$$|\psi_S(t)\rangle = \hat{U}(t, 0) |\psi_0\rangle, \quad \hat{O}_S(t) = \hat{O} \quad (1.84)$$

In the Heisenberg picture, on the other hand, the state vector remains a constant while the operator evolves with time:

$$|\psi_H(t)\rangle = |\psi_0\rangle, \quad \hat{O}_H(t) = \hat{U}^\dagger(t, 0) \hat{O} \hat{U}(t, 0) \quad (1.85)$$

It is straightforward to show that both pictures yield the same physical observable

$$\langle O(t) \rangle = \langle \psi_S(t) | \hat{O}_S(t) | \psi_S(t) \rangle = \langle \psi_H(t) | \hat{O}_H(t) | \psi_H(t) \rangle = \langle \psi_0 | \hat{U}^\dagger(t, 0) \hat{O} \hat{U}(t, 0) | \psi_0 \rangle \quad (1.86)$$

In general, it is more convenient to use the Heisenberg picture to compute expectation values of an observable, while the Schrodinger picture is more convenient to extract time dependence of the state vector and its related properties, such as entanglement entropy.

1.5.2 *Quantum Zeno effect

Consider a quantum system under repeated measurements of a physical observable $\hat{O}$ with the following spectral decomposition:

$$\hat{O} = \sum_q q |q\rangle\langle q| \quad (1.87)$$

Without loss of generality, let's assume the initial state of the system is $\hat{O}$ eigenstate $|0\rangle$ with eigenvalue q_0 .

Now consider a generic Hamiltonian

$$\hat{H} = H_{0,0}|0\rangle\langle 0| + \sum_{q \neq 0} (H_{q,0}|q\rangle\langle 0| + h.c.) + \sum_{p,q \neq 0} H_{p,q}|p\rangle\langle q| \quad (1.88)$$

that governs the time evolution of the system. We used *h.c.* as a shorthand notation for “hermitian conjugate”.

Starting from the initial state $|0\rangle$, after a very short time Δt , the state evolves into

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|0\rangle = (1 - \frac{i}{\hbar}H_{0,0}\Delta t)|0\rangle - \frac{i}{\hbar}\Delta t \sum_q H_{q,0}|q\rangle + O(\Delta t^2) \quad (1.89)$$

Therefore at time Δt , the probability of measurement outcome $\hat{O} = q_0$ is given by

$$|\langle 0 | \psi(t) \rangle|^2 = 1 - \alpha(\Delta t)^2 + O(\Delta t^3), \quad \alpha > 0 \quad (1.90)$$

where the constant α depends on the Hamiltonian $\hat{H}$.

One can further show that if we repeat this 2-step procedure of small time evolution for $\Delta t = \frac{T}{N}$ followed by a measurement of operator $\hat{O}$, after N repetitions, the probability for the state to come back to itself approaches 1 in the large N limit:

$$\Pr(|0\rangle \rightarrow |0\rangle) \geq \lim_{N \rightarrow \infty} (|\langle 0 | \psi(\frac{T}{N}) \rangle|^2)^N = \lim_{N \rightarrow +\infty} e^{-\alpha T^2/N} = 1 \quad (1.91)$$

This is known as the quantum Zeno effect.

1.6 Composite quantum systems (9/11)

- Refer to N&C 2.1.7, 2.2.8, 4.3; Tong 3.6. For generalized measurements, refer to N&C 2.2.6 and Tong 16.4.2

1.6.1 Tensor product

Previously we have focused on a single closed quantum system. When two closed quantum systems are combined together e.g. through some interactions b/w the two systems, how to describe such as composite quantum system? The last postulate addresses this issue, under the mathematical frameworks of *tensor product*:

Postulate 1.11. *The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n , and the system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$.*

Specifically, consider two quantum systems with Hilbert space V of dimension m and W of dimension n , their composite system is described by the tensor product of the two Hilbert spaces:

$$V \otimes W \equiv \{|v\rangle \otimes |w\rangle | |v\rangle \in V, |w\rangle \in W\} \quad (1.92)$$

which is a Hilbert space of dimension mn . Linear operators on $V \otimes W$ are given by $A \otimes B$, where A is a linear operator on V and B is a linear operator on W , satisfying

$$(A \otimes B)|v\rangle \otimes |w\rangle = A|v\rangle \otimes B|w\rangle \quad (1.93)$$

An inner product on $V \otimes W$ is naturally defined as

$$\left(\sum_i a_i |v_i\rangle \otimes |w_i\rangle, \sum_j b_j |v'_j\rangle \otimes |w'_j\rangle \right) = \sum_{i,j} a_i^* b_j \langle v_i | v'_j \rangle \langle w_i | w'_j \rangle \quad (1.94)$$

One can therefore define a complete set of orthonormal basis $\{|v_i\rangle \otimes |w_j\rangle\}$ for $V \otimes W$, and the matrix representation of linear operators in $V \otimes W$ is given by

$$(A \otimes B)_{i,j} = A_{i_1,j_1} B_{i_2,j_2}, \quad A \in L_V, \quad B \in L_W. \quad (1.95)$$

Since $A \otimes B$ is a matrix of dimension mn , where m and n are dimensions of matrices A and B respectively, the row and column index of matrix $A \otimes B$ can be written as

$$i = (i_1 - 1)n + i_2, \quad j = (j_1 - 1)n + j_2. \quad (1.96)$$

More properties of tensor product can be found in N&C section 2.1.7.

A *product state* in a composite system $\mathbb{V}_1 \otimes \mathbb{V}_2$ is defined as

$$|\psi\rangle_{12} = |\psi_1\rangle \otimes |\psi_2\rangle \quad (1.97)$$

It has NO quantum entanglement between subsystems 1 and 2.

With the help of tensor product, we can define Hamiltonian and unitary operators for a composite quantum system. We will study the transverse field Ising model for a 2-qubit system

$$H_{\text{TFIM}} = -JX_1 \otimes X_2 - h(Z_1 \otimes \hat{1}_2 + \hat{1}_2 \otimes Z_2) \quad (1.98)$$

as a demonstration. Due to a conserved quantity $Z_1 \otimes Z_2$ that commutes with the Hamiltonian, it can be diagonalized in the $Z_1 \otimes Z_2 = \pm 1$ subspaces respectively.

Name	Controlled-Z (CPHASE)	CNOT	SWAP
Unitary matrix	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Table 1.2: Common 2-qubit gates and associated unitary matrices.

Name	Toffoli (Controlled-CNOT)	Fredkin (Controlled-SWAP)
Unitary matrix	$\frac{1+Z_1}{2} \otimes \text{CNOT}_{23} + \frac{1-Z_1}{2} \otimes \hat{1}_{23}$	$\frac{1+Z_1}{2} \otimes \text{SWAP}_{23} + \frac{1-Z_1}{2} \otimes \hat{1}_{23}$

Table 1.3: Common 3-qubit gates and associated unitary matrices.

A good set of examples of unitary operators in a composite system are 2-qubit and 3-qubit quantum gates, shown in Table 1.2 and 1.3 respectively, such as a controlled- U gate in the 2-qubit system.

1.6.2 Generalized measurements

Composite systems play an important role in understanding general measurements in quantum mechanics, where we consider the measured system (Hilbert space $\mathbb{V}_s$) and an ancilla (Hilbert space $\mathbb{V}_a$) together as a composite quantum system, whose Hilbert space is the tensor product $\mathbb{V}_s \otimes \mathbb{V}_a$.

Consider an initial state

$$|\Phi\rangle \equiv |\psi\rangle_s \otimes |\phi\rangle_a \quad (1.99)$$

unitarily evolving under a Hamiltonian $\hat{H}$ of the composite system for time t . Next we perform a projective measurement on the ancilla, associated with observable $\hat{1}_s \otimes \hat{O}_a$, where Hermitian operator $\hat{O}$ has the following spectral decomposition:

$$\hat{O} = \sum_q q |q\rangle\langle q| \quad (1.100)$$

The basis vectors $\{|q\rangle\}$ forms a complete set of orthonormal basis for ancilla Hilbert space $\mathbb{V}_a$.

After this measurement, if the measurement outcome is q , the system collapses into a new state $M_q|\psi\rangle_s \otimes |q\rangle_a$ (unnormalized), where linear operator M_q is defined as

$$M_q \equiv \langle q|_a e^{-i\hat{H}t/\hbar} |\phi\rangle_a \in L_{\mathbb{V}_s} \quad (1.101)$$

satisfying the completeness relation

$$\sum_q M_q^\dagger M_q = \hat{1}_s \quad (1.102)$$

The probability for measurement outcome q is given by

$$\text{Pr}(q) = \text{Tr}(M_q|\psi\rangle_s\langle\psi|_s M_q^\dagger) = \langle\psi|M_q^\dagger M_q|\psi\rangle \quad (1.103)$$

Clearly, the completeness relation (1.102) for operators $\{M_q\}$ guarantees the normalization of

probability distribution:

$$\sum_q \text{Pr}(q) = 1. \quad (1.104)$$

If we disregard the ancilla and examine this measurement from the viewpoint of the system, for measurement outcome q , initial state $|\psi\rangle$ becomes $M_q|\psi\rangle$, with a probability $\text{Pr}(q)$ given in (1.103). This is known as a generalized measurement.

If the Kraus operator M_q is very close to identity matrix $\hat{1}_s$

$$M_q = \sqrt{1 - \lambda_q} \hat{1} + \hat{O}_q, \quad \lambda_q \ll 1 \quad (1.105)$$

such a generalized measurement is known as a weak measurement because the measurement only modifies the initial state weakly in the sense that

$$\left| |\psi\rangle - M_q|\psi\rangle \right| \ll 1 \quad (1.106)$$

This is the case when initial state $|\phi\rangle = |q\rangle$ and the evolution time t is much smaller than any energy scale in Hamiltonian $\hat{H}$.

2 Density matrix formulation of quantum mechanics

2.1 Density matrix and mixed states (9/16)

- Refer to N&C 2.4, 2.5, 11.3; Tong 16.3

2.1.1 The density matrix

If we think of a physical system (with Hilbert space A) and its environment (with Hilbert space B) combined together as a (composite) closed quantum system, any state therein is described by a state vector

$$|\psi\rangle = \sum_{i,j} \psi_{i,j} |a_i\rangle \otimes |b_j\rangle \quad (2.1)$$

where $\{|a_i\rangle | 1 \leq i \leq m = \dim A\}$ is a complete set of orthonormal basis for A , while $\{|b_j\rangle | 1 \leq j \leq n = \dim B\}$ is a complete set of orthonormal basis for B .

To study physical properties of system A , the composite state $|\psi\rangle$ in (2.1) is an overkill because it also contains all information about the environment, in which we are not interested at all. It turns out the following quantity, known as the *density matrix* (or *density operator*) for subsystem A

$$\hat{\rho} = \text{Tr}_B |\psi\rangle\langle\psi| = \sum_{i_1, i_2=1}^m \rho_{i_1, i_2} |a_{i_1}\rangle\langle a_{i_2}|, \quad \rho_{i_1, i_2} = \sum_{j=1}^n \psi_{i_1, j} \psi_{i_2, j}^* \quad (2.2)$$

encodes all information about physical system A in composite state $|\psi\rangle$. As shown above it is obtained by tracing out subsystem B in state $|\psi\rangle$. Any physical observable of the system A corresponds to a Hermitian operator $\hat{O}_A \times \hat{1}_B$ in the composite system $A \times B$, and its expectation value is nothing but

$$\langle \hat{O} \rangle_A \equiv \langle \psi | \hat{O}_A \otimes \hat{1}_B | \psi \rangle = \text{Tr}(\hat{\rho} \hat{O}) \quad (2.3)$$

Instead of directly computing the partial trace, there is another efficient way to obtain the density matrix (2.2), known as *Schmidt decomposition*, which we describe below. This approach is especially informative for understanding entanglement between subsystem A and B .

We can treat the wavefunction $\psi_{i,j}$ in (2.1) as a matrix of dimension $m \times n$, which has the following singular value decomposition (SVD):

$$\psi = U^\dagger D V, \quad U^\dagger U = V^\dagger V = \hat{1}, \quad D_{i,j} = \delta_{i,j} \lambda_i, \quad \lambda_i \geq 0. \quad (2.4)$$

where U, V are two unitary matrices of dimensions $m \times m$ and $n \times n$ respectively. D is a $m \times n$ matrix whose matrix elements all vanish except for positive diagonal elements $\{\lambda_i > 0 | 1 \leq i \leq r, r \leq \min(m, n)\}$, known as singular values of matrix ψ . The positive integer $r \leq \min(m, n)$, is the rank of matrix $\hat{\psi}$.

Since unitary matrices implement the change of basis, the SVD procedure introduced a new set of orthonormal basis (known as the Schmidt basis) for both the system A and environment B :

$$|A_i\rangle = \sum_{i'} U_{i,i'}^* |a_{i'}\rangle, \quad |B_j\rangle = \sum_{j'} V_{j,j'} |b_{j'}\rangle \quad (2.5)$$

such that the state vector (2.1) can be written as

$$|\psi\rangle = \sum_{i=1}^r \lambda_i |A_i\rangle \otimes |B_i\rangle, \quad \lambda_i > 0 \quad (2.6)$$

This is known as the Schmidt decomposition of state vector $|\psi\rangle$ in a composite Hilbert space, which is the tensor product of A and B . The rank r of matrix ψ is known as the Schmidt rank (or Schmidt number), and the positive numbers $\{\lambda_i > 0 | 1 \leq i \leq r\}$ are known as Schmidt coefficients.

Note that normalization of state vector $|\psi\rangle$ implies that $\{p_i \equiv \lambda_i^2\}$ leads to a normalized probability distribution:

$$\langle\psi|\psi\rangle = \sum_{i=1}^r \lambda_i^2 = 1 = \sum_{i=1}^r p_i \quad (2.7)$$

In the Schmidt basis $\{|A_i\rangle\}$, the density matrix (2.2) has the following diagonal form:

$$\hat{\rho} = \sum_{i=1}^r \lambda_i^2 |A_i\rangle\langle A_i| = \sum_{i=1}^r p_i |A_i\rangle\langle A_i| \quad (2.8)$$

which has the physical meaning as an ensemble $\{p_i, |A_i\rangle\}$ of pure states. p_i is the probability for the system A to fall into pure state $|A_i\rangle$.

Mathematically, a density matrix is defined as a linear operator $\hat{\rho}$ satisfying 2 conditions:

(1) **Trace condition**

$$\text{Tr } \hat{\rho} = 1 \quad (2.9)$$

(2) **Positivity condition**

$$\hat{\rho} \geq 0 \quad (2.10)$$

The positivity of $\hat{\rho}$ already implies that it is a Hermitian operator i.e. $\hat{\rho} = \hat{\rho}^\dagger$.

All density matrices form a convex set in the sense that $x\rho_1 + (1-x)\rho_2$, $\forall 0 \leq x \leq 1$ is also a density matrix if $\rho_{1,2}$ are density matrices.

In the diagonal basis (2.8) of density matrix $\hat{\rho}$, if the Schmidt rank $r > 1$ i.e. there are at least two independent pure states in the ensemble, we call $\hat{\rho}$ a *mixed state*. If $r = 1$, instead we call $\hat{\rho} = |A_1\rangle\langle A_1|$ to be a *pure state*.

Theorem 2.1. *A density matrix $\hat{\rho}$ corresponds to a pure state if and only if*

$$\hat{\rho}^2 = \hat{\rho} \quad (2.11)$$

As a simplest example of Schmidt decomposition, we consider a single-qubit physical system A and a single-qubit environment B , where a pure state of the combined closed system has the following general form:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle \quad (2.12)$$

where we have used $|0\rangle \equiv |\uparrow\rangle$ and $|1\rangle \equiv |\downarrow\rangle$ (known as the computational basis) to label eigenstates of the Pauli- $\hat{Z}$ operator. We also used a shorthand notation for tensor product states:

$$|ab\rangle \equiv |a\rangle_A \otimes |b\rangle_B, \quad a, b = 0, 1. \quad (2.13)$$

In this case we have the matrix $\psi_{a,b}$ as

$$\hat{\psi} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad (2.14)$$

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for the SVD procedure.

Similar to the case of pure states, in a composite system $\mathbb{V}_1 \otimes \mathbb{V}_2$, one can also define a *product state* for generic mixed states:

$$\rho_{\text{product}} = \rho_1 \otimes \rho_2 \quad (2.15)$$

and a *separate state*:

$$\rho_{\text{separable}} = \sum_i p_i \rho_{1,i} \otimes \rho_{2,i}, \quad \sum_i p_i = 1. \quad (2.16)$$

A product state features vanishing mutual information (see next section) between subsystems 1 and 2, while a separable state features vanishing quantum entanglement (e.g. in terms of entanglement negativity, see HW4) between subsystems 1 and 2.

2.1.2 Entanglement entropy

In the Schmidt decomposition described previously, if the density matrix $\hat{\rho}$ is obtained by a partial trace over B from state ρ_{AB} in the composite system $A \otimes B$:

$$\hat{\rho}_A = \text{Tr}_B \hat{\rho}_{AB} \quad (2.17)$$

it is called a *reduced density matrix*. Importantly, the reduced density matrix characterizes the correlation and entanglement between subsystem A and B , as we illustrate below.

Consider the following Hamiltonian in a composite system of two qubits A and B :

$$H = -JX_A \otimes X_B - hZ_A \otimes Z_B, \quad h, J > 0 \quad (2.18)$$

It can be simultaneously diagonalized with the conserved quantities $Z_A \otimes Z_B$ and $X_A \otimes X_B$. Its ground state with the lowest energy $E_0 = -h - J$ is given by

$$|Bell_0\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (2.19)$$

while the 3 excited states are given by

$$|Bell_1\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad (2.20)$$

$$|Bell_2\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \quad (2.21)$$

$$|Bell_3\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}. \quad (2.22)$$

These 4 states are known as *Bell states* or *EPR pairs*. They are all examples of bipartite *maximally entangled states* in the two-qubit system, in the sense that their reduced density matrix is proportional to the identity operator:

$$\hat{\rho}_A = \text{Tr}_B |Bell_j\rangle\langle Bell_j| = \frac{1}{2}\hat{1}_A, \quad j = 0, 1, 2, 3. \quad (2.23)$$

Generally, a bipartite pure state $|\psi_{AB}\rangle$ is maximally entangled if and only if

$$\hat{\rho}_A \equiv \text{Tr}_B |\psi_{AB}\rangle\langle\psi_{AB}| = \frac{1}{\dim A} \hat{1}_A \quad (2.24)$$

In order to quantitatively measure the quantum entanglement between two subsystems, we

need to introduce the concept of von Neumann entropy for a density matrix $\hat{\rho}$:

$$S(\hat{\rho}) \equiv -\text{Tr } \hat{\rho} \log \hat{\rho} \quad (2.25)$$

In the diagonal form (2.8) of density matrix $\hat{\rho}$, the von Neumann entropy defined above reduces to the Shannon entropy

$$S = -\sum_j p_j \log p_j \quad (2.26)$$

for a probability distribution $\{p_j\}$, an important concept in information theory. If we take the equilibrium density matrix $\hat{\rho} \propto e^{-\hat{H}/k_B T}$, the von Neumann entropy is nothing but the thermodynamic entropy of the corresponding quantum system described by Hamiltonian $\hat{H}$.

Now for a pure state $|\psi_{AB}\rangle$ in our bipartite system $A \otimes B$, if we consider the reduced density matrix $\hat{\rho}_A$ for subsystem A , its associated von Neumann entropy is called the *entanglement entropy*, because it quantitatively captures the entanglement (or more precisely, correlation) between subsystems A and B . The entanglement entropy has the following properties:

- (1) Positivity: $S(\hat{\rho}) \geq 0$. In particular, $S(\hat{\rho}) = 0$ if and only if $\hat{\rho}$ is a pure state.
- (2) Upper bound: $S(\hat{\rho}_A) \leq \log(\dim A)$, which is saturated by $\hat{\rho}_A = \hat{1}_A/\dim A$. This is why (2.24) is called a maximally entangled state.
- (3) $S(\hat{\rho}_A) = S(\hat{\rho}_B)$, which comes directly from Schmidt decomposition of pure state $|\psi_{AB}\rangle$. If $S(\hat{\rho}_A) = S(\hat{\rho}_B) > 0$, we say that A and B subsystems are (bipartite) entangled in a pure state $|\psi_{AB}\rangle$.
- (4) Subadditivity: $S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B)$ for a mixed state ρ_{AB} . The equality holds if and only if $\rho_{AB} = \rho_A \otimes \rho_B$, which is called a bipartite *product state*. This property allows for defining the *mutual information* between subsystems A and B

$$I(A : B) \equiv S(\rho_A) + S(\rho_B) - S(\rho_{AB}) \geq 0 \quad (2.27)$$

which is a measure of entanglement (or quantum correlations) between A and B in a mixed state. The mutual information reduces to twice the entanglement entropy for a pure state.

- (5) Araki-Lieb inequality: $S(\rho_{AB}) \geq |S(\rho_A) - S(\rho_B)|$.
- (6) Concavity: $S(\sum_i p_i \rho_i) \geq \sum_i p_i S(\rho_i)$ for any probability distribution $\{p_i\}$ and any density matrices $\{\rho_i\}$.
- (7) Strong subadditivity: refer to N&C 11.4.

2.2 Witnessing quantum entanglement (9/18)

- Refer to Tong 16.1.1-16.1.4, S&Z Chapter 18

2.2.1 The EPR “paradox”

Einstein, Podolsky and Rosen (EPR) investigated the singlet Bell state of 2 qubits

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (2.28)$$

which was also known as the EPR state. Assuming that Alice has qubit 1 and Bob has qubit 2, they observed that a measurement of Pauli $\hat{Z}_1$ operator by Alice (say, with outcome $q = \pm 1$) will instantaneously collapse the EPR state into a product state $|q\rangle_1 \otimes |-q\rangle_2$, and immediately determine the measurement outcome of Pauli $\hat{Z}_2$ operator by Bob, no matter how far away Alice and Bob are physically separated.

EPR considered the following 3 claims which all seemed reasonable, from the perspective of classical physics:

Locality: Alice’s choice cannot instantaneously alter the physical reality at Bob’s distant location.

Reality criterion: Perfectly predictable results must reflect preexisting properties.

Quantum completeness: The wavefunction completely describes physical reality.

EPR concluded that these three claims cannot all hold simultaneously. Because they retained locality and the reality criterion, they concluded that the quantum state is incomplete, perhaps omitting hidden variables that specify the missing spin values.

The Einstein–Podolsky–Rosen paradox asks whether quantum mechanics can be replaced by a theory that is both local and realistic. Consider two spatially separated observers, Alice and Bob, who receive qubits 1 and 2. Alice may measure either Q or R , while Bob may measure either S or T . Every measurement has outcomes in $\{+1, -1\}$:

$$Q^2 = R^2 = S^2 = T^2 = 1. \quad (2.29)$$

Measurements associated with different parties are compatible, for example $[Q, S] = 0$.

In a local realistic model, measurement outcomes are assumed to have pre-existing values and Alice’s choice does not affect Bob’s value. Thus, for a hidden-variable assignment,

$$E(QS) = E(Q)E(S), \quad (2.30)$$

where each individual value $E(Q), E(R), E(S), E(T)$ is ± 1 .

2.2.2 The CHSH inequality

Define the CHSH combination

$$\mathcal{B} = QS + RS + RT - QT. \quad (2.31)$$

For a local realistic assignment,

$$E(\mathcal{B}) = [E(Q) + E(R)]E(S) + [E(R) - E(Q)]E(T). \quad (2.32)$$

Since $E(R) = E(Q)$ or $E(R) = -E(Q)$, precisely one bracket vanishes and the other equals ± 2 . Consequently,

$$\boxed{|E(QS + RS + RT - QT)| \leq 2.} \quad (2.33)$$

This is the Clauser–Horne–Shimony–Holt (CHSH) inequality.

2.2.2.1 Quantum-mechanical violation

Let the observables be spin measurements along unit vectors $\mathbf{m}_{1,2}$ and $\mathbf{n}_{1,2}$:

$$Q = \mathbf{m}_1 \cdot \boldsymbol{\sigma}_1, \quad R = \mathbf{n}_1 \cdot \boldsymbol{\sigma}_1, \quad S = \mathbf{m}_2 \cdot \boldsymbol{\sigma}_2, \quad T = \mathbf{n}_2 \cdot \boldsymbol{\sigma}_2. \quad (2.34)$$

The two qubits are prepared in the singlet Bell state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle). \quad (2.35)$$

For this state,

$$\langle \Psi^- | (\mathbf{a} \cdot \boldsymbol{\sigma}_1)(\mathbf{b} \cdot \boldsymbol{\sigma}_2) | \Psi^- \rangle = -\mathbf{a} \cdot \mathbf{b}. \quad (2.36)$$

Hence

$$\langle \mathcal{B} \rangle = -\mathbf{m}_1 \cdot \mathbf{m}_2 - \mathbf{n}_1 \cdot \mathbf{m}_2 - \mathbf{n}_1 \cdot \mathbf{n}_2 + \mathbf{m}_1 \cdot \mathbf{n}_2 \quad (2.37)$$

$$= -(\mathbf{m}_1 + \mathbf{n}_1) \cdot \mathbf{m}_2 + (\mathbf{m}_1 - \mathbf{n}_1) \cdot \mathbf{n}_2. \quad (2.38)$$

Choose coplanar directions with neighboring axes separated by 45° . Then

$$|\langle \mathcal{B} \rangle| = 2\sqrt{2} > 2. \quad (2.39)$$

The value $2\sqrt{2}$ is the maximal quantum value, known as the Tsirelson bound.

2.2.2.2 Experimental test with photon polarization

In an optical realization, a source produces pairs of entangled photons. One photon is sent to polarimeter A and the other to polarimeter B . Each polarimeter selects a measurement orientation, and a coincidence monitor records joint detection events. With

$$|V\rangle = \text{vertical polarization}, \quad |H\rangle = \text{horizontal polarization}, \quad (2.40)$$

polarization measurements implement the two-outcome observables required for a Bell test.

2.2.3 *The GHZ argument

Consider the three-qubit GHZ state

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}} (|000\rangle - |111\rangle). \quad (2.41)$$

The state obeys

$$X_A Y_B Y_C |\text{GHZ}\rangle = +|\text{GHZ}\rangle, \quad (2.42)$$

$$Y_A X_B Y_C |\text{GHZ}\rangle = +|\text{GHZ}\rangle, \quad (2.43)$$

$$Y_A Y_B X_C |\text{GHZ}\rangle = +|\text{GHZ}\rangle, \quad (2.44)$$

and

$$X_A X_B X_C |\text{GHZ}\rangle = -|\text{GHZ}\rangle. \quad (2.45)$$

Suppose a local hidden-variable theory assigns predetermined values $s_j^{(x)}, s_j^{(y)} \in \{+1, -1\}$ to

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the X and Y measurements on qubit $j = A, B, C$. The first three GHZ correlations require

$$s_A^{(x)} s_B^{(y)} s_C^{(y)} = +1, \quad (2.46)$$

$$s_A^{(y)} s_B^{(x)} s_C^{(y)} = +1, \quad (2.47)$$

$$s_A^{(y)} s_B^{(y)} s_C^{(x)} = +1. \quad (2.48)$$

Multiplying these equations gives

$$s_A^{(x)} s_B^{(x)} s_C^{(x)} = +1. \quad (2.49)$$

Quantum mechanics instead predicts a product of -1 . This is a direct contradiction, with no statistical inequality required.

2.2.4 *Bell's Original Inequality

Return to the singlet state

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle). \quad (2.50)$$

For a unit vector $\mathbf{n}$,

$$\mathbf{n} \cdot \boldsymbol{\sigma} = n_x \sigma_x + n_y \sigma_y + n_z \sigma_z, \quad |\mathbf{n}| = 1, \quad (2.51)$$

and

$$\mathbf{n} \cdot \boldsymbol{\sigma} |\pm \mathbf{n}\rangle = \pm |\pm \mathbf{n}\rangle. \quad (2.52)$$

In any common spin basis,

$$|\text{EPR}\rangle = \frac{1}{\sqrt{2}} (|+\mathbf{n}, -\mathbf{n}\rangle - |-\mathbf{n}, +\mathbf{n}\rangle). \quad (2.53)$$

Consider measurement axes $\mathbf{n}_a$, $\mathbf{n}_b$, and $\mathbf{n}_c$. Let P_{ab} denote the probability that the first qubit has outcome $+1$ along $\mathbf{n}_a$ while the second has outcome $+1$ along $\mathbf{n}_b$. Writing $p(s_a s_b s_c)$ for a predetermined hidden-variable assignment,

$$P_{ab} = p(+, -, +) + p(+, -, -), \quad (2.54)$$

$$P_{bc} = p(+, +, -) + p(-, +, -), \quad (2.55)$$

$$P_{ac} = p(+, -, -) + p(-, +, -). \quad (2.56)$$

Nonnegativity of probabilities implies a three-axis probability inequality

$$\boxed{P_{ab} + P_{bc} \geq P_{ac}.} \quad (2.57)$$

This inequality is indeed violated by quantum probabilities. For coplanar directions, parameterize the spin eigenstates by

$$|+\theta\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix}, \quad |-\theta\rangle = \begin{pmatrix} -\sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}. \quad (2.58)$$

For the singlet,

$$P_{ij} = \frac{1}{2} \sin^2 \left(\frac{\theta_i - \theta_j}{2} \right). \quad (2.59)$$

Choose

$$\theta_a = 0, \quad \theta_b = \theta, \quad \theta_c = 2\theta. \quad (2.60)$$

Then

$$P_{ab} = P_{bc} = \frac{1}{2} \sin^2\left(\frac{\theta}{2}\right), \quad P_{ac} = \frac{1}{2} \sin^2 \theta. \quad (2.61)$$

Bell's inequality would require

$$\sin^2\left(\frac{\theta}{2}\right) \geq \frac{1}{2} \sin^2 \theta. \quad (2.62)$$

For $\theta = 60^\circ$,

$$\frac{1}{4} \not\geq \frac{3}{8}, \quad (2.63)$$

which contradicts the local hidden-variable prediction.

2.2.5 Summary

Below is a summary of these entanglement witnesses:

- Local realism leads to the CHSH bound $|\langle \mathcal{B} \rangle| \leq 2$.
- The singlet state can reach $2\sqrt{2}$, violating the CHSH inequality.
- GHZ correlations give a deterministic contradiction with predetermined local outcomes.
- Bell's original probability inequality is also violated by singlet correlations for suitable measurement directions.

2.3 *Generalized measurements and decoherence (9/23)

- Refer to N&C 2.2.6, 2.4.2, 8.2; Tong 16.4; Anastopoulos 8.2

2.3.1 Generalized measurements on a mixed state

Consider the product state $\rho_s \otimes \rho_a$ as the initial state of a composite system $\mathbb{V}_s \otimes \mathbb{V}_a$, consisting of a physical system s unentangled with the experimental apparatus a . We assume the apparatus density matrix ρ_a has the following spectral decomposition:

$$\rho_a = \sum_j p_j |j\rangle_a \langle j|_a, \quad \sum_j p_j = 1, \quad 0 < p_j \leq 1. \quad (2.64)$$

where $\{|j\rangle_e\}$ is a complete set of orthonormal basis for the apparatus Hilbert space $\mathbb{V}_e$.

The system and apparatus evolve together under a unitary operator $\hat{U}$, before a projective measurement is performed for an observable (i.e. Hermitian operator) on the apparatus Hilbert space

$$\hat{O} = \hat{1}_s \otimes \left(\sum_q q |q\rangle_a \langle q|_a \right) \quad (2.65)$$

It's straightforward to show that for measurement outcome q , the initial state ρ_s of the physical system collapses into the following (unnormalized) state

$$\rho_q \propto \sum_j M_{q,j} \rho_s M_{q,j}^\dagger \quad (2.66)$$

with a probability distribution

$$\Pr(q) = \sum_j \text{Tr}(M_{q,j} \rho_s M_{q,j}^\dagger) \quad (2.67)$$

where the measurement operators $\{M_{q,j}\}$ are defined as

$$M_{q,j} \equiv \langle q | \hat{U} | j \rangle \sqrt{p_j} \quad (2.68)$$

satisfying the completeness relation

$$\sum_{q,j} M_{q,j}^\dagger M_{q,j} = \hat{1}_s \quad (2.69)$$

The above discussion is completely in parallel to section 1.6.2, merely generalizing everything to the context of mixed states. Therefore we can formulate the measurement postulate of quantum mechanics for density matrices:

Postulate 2b: A *generalized quantum measurement* is described by a collection $\{M_m\}$ of measurement operators satisfying the completeness relation:

$$\sum_m M_m^\dagger M_m = \hat{1}. \quad (2.70)$$

If the state of the quantum system is ρ immediately before the measurements, then the probability that measurement result m occurs is given by

$$\Pr(m) = \text{Tr}(\rho M_m^\dagger M_m) \quad (2.71)$$

and the (unnormalized) state of the system after the measurement is $\rho_m \propto M_m \rho M_m^\dagger$.

In a generalized measurement, the set of positive operators $\{E_m \equiv M_m^\dagger M_m\}$ is known as a POVM (positive operator-value measure). A generalized measurement is sometimes also known as a POVM measurement.

2.3.2 Examples of generalized measurements

(1) Quantum Zeno effect

In a single-qubit system, consider a repeated 2-step procedure where the 1st step of unitary time evolution for a duration T/N is followed by the 2nd step of a projective measurement on $\hat{X}$:

$$\hat{X} = \hat{P}_+ - \hat{P}_-, \quad \hat{P}_+ = |+\rangle\langle+|, \quad \hat{P}_- = |-\rangle\langle-|. \quad (2.72)$$

where

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle) \quad (2.73)$$

are the eigenstates of Pauli $\hat{X}$ operator.

In this example, the measurement operators associated with the $\pm$ outcome after each step are $\{\hat{M}_\pm\}$ defined as follows:

$$\hat{M}_\pm = \hat{P}_\pm U(T/N), \quad M_+^\dagger M_+ + M_-^\dagger M_- = \hat{1}. \quad (2.74)$$

(2) Detection of a photon

Let $|0\rangle$ denote no photon and $|1\rangle$ denote one photon. A simple photon detection model uses

$$\hat{M}_0 = |0\rangle\langle 0|, \quad \hat{M}_c = |0\rangle\langle 1| \quad (2.75)$$

where the subscript c denotes a click in the photon reader. The photon is annihilated after the click outcome. It's straightforward to check the completeness relation holds for $\{M_0, M_c\}$.

(3) Distinguishing quantum states

Suppose a qubit is prepared in either

$$|0\rangle \quad \text{or} \quad |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

The task is to determine which state was received.

(i) Projective measurement in the Z basis:

The measurement operators are projection operators

$$\hat{P}_0 = |0\rangle\langle 0|, \quad \hat{P}_1 = |1\rangle\langle 1|.$$

If the outcome is 1, the received state must have been $|+\rangle$ because $\langle 1|0\rangle = 0$. If the outcome is 0, the two possibilities cannot be distinguished.

(ii) Projective measurement in the X basis:

The measurement operators are again projectors

$$\hat{P}_+ = |+\rangle\langle+|, \quad \hat{P}_- = |-\rangle\langle-|.$$

If the outcome is $+$, the states cannot be distinguished. If the outcome is $-$, the received state must have been $|0\rangle$ because $\langle -|+\rangle = 0$.

(iii) A three-outcome POVM:

Consider the POVM $\{E_m = M_m^\dagger M_m\}$:

$$\begin{aligned}\hat{E}_1 &= \frac{1}{2}|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \\ \hat{E}_2 &= \frac{1}{2}|-\rangle\langle -| = \frac{1}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \\ \hat{E}_3 &= \hat{1} - \hat{E}_1 - \hat{E}_2 = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix} \geq 0.\end{aligned}$$

The outcomes have the following interpretation:

1. Outcome 1: the state must have been $|+\rangle$.
2. Outcome 2: the state must have been $|0\rangle$.
3. Outcome 3: the measurement is inconclusive.

2.3.3 Measurements induce decoherence

In its eigenbasis $\{|j\rangle | 1 \leq j \leq \dim \mathbb{V}\}$, a mixed state (2.8) is an ensemble of pure states $\{|j\rangle\}$

$$\hat{\rho} = \sum_j p_j |j\rangle\langle j|, \quad \sum_j p_j = 1. \quad (2.76)$$

while a generic pure state in the same physical system can be written as a (normalized) superposition of the basis vectors $\{|j\rangle\}$:

$$|\psi\rangle = \sum_j \psi_j |j\rangle, \quad \sum_j |\psi_j|^2 = 1. \quad (2.77)$$

But there is a big difference between the (“incoherent”) ensemble summation in the mixed state (2.76) and the (“coherent”) superposition in the pure state (2.77), in the sense that the pure state contains more information in the relative phases of coefficients ψ_j that can be measured experimentally. To make it clear, we use the normalization condition in (2.77) to define the probability distribution $p_j \equiv |\psi_j|^2$, and the normalized pure state can be rewritten as

$$|\psi\rangle = \sum_j \sqrt{p_j} e^{i\phi_j} |j\rangle, \quad \sum_j p_j = 1. \quad (2.78)$$

While the global phase factor is a redundancy with no physical significance, there are $(\dim \mathbb{V} - 1)$ relative phases that can be detected by physical observables.

To illustrate, consider a single qubit example, where we use $(\hat{X}, \hat{Y}, \hat{Z})$ to label the 3 Pauli matrices. We consider a mixed state $\hat{\rho}$ and a pure state ψ , both having 1/2 probability for either $Z = +1$ or $Z = -1$. In the computational basis $|0\rangle \equiv |\uparrow\rangle$, $|1\rangle \equiv |\downarrow\rangle$ that diagonalizes $\hat{Z}$, the mixed state is written as

$$\hat{\rho} = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (2.79)$$

and the pure state as

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\phi}|1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ e^{i\phi} \end{pmatrix} \quad (2.80)$$

If we now measure another observable $\hat{X}$ for both states, we end up with different expectation values:

$$\text{Tr}(\hat{\rho}\hat{X}) = 0, \quad \langle\psi|\hat{X}|\psi\rangle = \cos\phi. \quad (2.81)$$

exactly rooted in the relative phases and associated quantum interferences in the pure state.

As a result, the reduction from a pure state (2.78) to a mixed state (2.76) is a process of decoherence, i.e. a reduction of quantum interference. This can be achieved by a projective measurement (see postulate 1.9), where the measurement outcome is unknown. Consider the implementation of a measurement for physical observable diagonal in basis $\{|j\rangle\}$

$$\hat{O} \equiv \sum_j O_j |j\rangle\langle j| \quad (2.82)$$

on the pure state (2.78), If we don't know the outcome, after a projective measurement, we should view the resulting state as a summation of projector $|j\rangle\langle j|$ weighted by probability $p_j = |\langle j|\psi\rangle|^2$, and hence we end up with the mixed state (2.76). This means a quantum measurement will generally decohere a quantum state, even if we don't know the measurement outcome. This is in sharp contrast to a classical system.

2.4 *Time evolution of open quantum systems (9/25)

- Refer to Tong 16.5.1-16.5.3, N&C 8.3

For a closed quantum system, its unitary time evolution can be easily reformulated using density matrices:

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle, \quad |\psi(t)\rangle = U(t, 0) |\psi(0)\rangle, \quad (2.83)$$

$$i\hbar \frac{d\rho}{dt} = [H, \rho], \quad \rho(t) = U(t, 0) \rho(0) U^\dagger(t, 0). \quad (2.84)$$

However this is not the most general time evolution for a mixed state $\rho(t)$ in an open quantum system, as we will show below.

2.4.1 Quantum channels and Kraus operators

Let system A interact with an environment B . Assume initially the environment is in a pure state $|0\rangle_B$:

$$\rho_{AB}(t=0) = \rho_A(0) \otimes \rho_B(0), \quad \rho_B(0) = |0\rangle\langle 0|_B. \quad (2.85)$$

After joint unitary evolution,

$$\rho_{AB}(t) = U(t, 0) (\rho_A(0) \otimes |0\rangle\langle 0|_B) U^\dagger(t, 0). \quad (2.86)$$

The reduced state in system A is

$$\rho_A(t) = \text{Tr}_B \rho_{AB}(t) \quad (2.87)$$

$$= \sum_k {}_B\langle k| U(t, 0) |0\rangle_B \rho_A(0) {}_B\langle 0| U^\dagger(t, 0) |k\rangle_B \quad (2.88)$$

$$= \sum_k M_k(t) \rho_A(0) M_k^\dagger(t), \quad (2.89)$$

where the *Kraus operators* $\{M_k\}$ defined by

$$M_k(t) = {}_B\langle k| U(t, 0) |0\rangle_B. \quad (2.90)$$

satisfy the completeness relation

$$\sum_k M_k^\dagger M_k = {}_B\langle 0| U^\dagger U |0\rangle_B = \hat{1}_A. \quad (2.91)$$

This indicates the time evolution of an open quantum system A is described by a quantum channel, or a CPTP map $\mathcal{E}$ defined below.

2.4.1.1 Definition of a quantum channel

A quantum channel is a completely positive, trace-preserving (CPTP) map, which can always be represented by a set of Kraus operators $\{M_m\}$:

$$\mathcal{E}(\rho) = \sum_m M_m \rho M_m^\dagger, \quad \sum_m M_m^\dagger M_m = \hat{1}. \quad (2.92)$$

It is trace preserving:

$$\text{Tr } \mathcal{E}(\rho) = \text{Tr } \rho \quad (2.93)$$

convex linear

$$\mathcal{E}(p\rho_1 + (1-p)\rho_2) = p\mathcal{E}(\rho_1) + (1-p)\mathcal{E}(\rho_2), \quad \forall 0 \leq p \leq 1. \quad (2.94)$$

and completely positive. Complete positivity means that for every auxiliary system B ,

$$(\mathcal{E}_A \otimes \mathcal{I}_B)(\rho_{AB}) \geq 0 \quad (2.95)$$

whenever $\rho_{AB} \geq 0$. A Kraus representation is not unique: two sets of Kraus operators $\{M_i\}$ and $\{N_j\}$ represent the same quantum channel if and only if they are related by a unitary matrix $u_{i,j}$:

$$M_i = \sum_j u_{i,j} N_j, \quad u^\dagger u = \hat{1}. \quad (2.96)$$

2.4.1.2 Example: a bit-flip channel

For two qubits with

$$H = -\hbar\omega \sigma_x^{(A)} \otimes \sigma_x^{(B)}, \quad (2.97)$$

the unitary is

$$U(t) = \cos(\omega t) \mathbb{I} + i \sin(\omega t) \sigma_x^{(A)} \otimes \sigma_x^{(B)}. \quad (2.98)$$

With the environment initially in $|0\rangle_B$,

$$M_0 = \cos(\omega t) \mathbb{I}_A, \quad M_1 = i \sin(\omega t) \sigma_x^{(A)}. \quad (2.99)$$

Therefore

$$\rho_A(t) = [1 - p(t)]\rho_A(0) + p(t)\sigma_x \rho_A(0) \sigma_x, \quad p(t) = \sin^2(\omega t). \quad (2.100)$$

Differentiation with respect to p gives

$$\frac{d\rho_A}{dp} = \sigma_x \rho_A \sigma_x - \rho_A, \quad \text{Tr}\left(\frac{d\rho_A}{dp}\right) = 0. \quad (2.101)$$

2.4.2 The Master Equation

2.4.2.1 Lindblad form

Markovian, time-local dynamics has the following Lindblad form

$$\boxed{\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_j \left(2L_j \rho L_j^\dagger - L_j^\dagger L_j \rho - \rho L_j^\dagger L_j \right)}. \quad (2.102)$$

The commutator with Hamiltonian $\hat{H}$ generates coherent evolution. The linear operators $\{L_j\}$ are known as *quantum jump operators*. The remaining terms describe dissipation and decoherence while preserving the trace.

2.4.2.2 Bloch-vector representation

A qubit state can be written as

$$\rho = \frac{1}{2}(\mathbb{I} + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad \mathbf{r} = (x, y, z), \quad |\mathbf{r}| \leq 1. \quad (2.103)$$

2 Density matrix formulation of quantum mechanics

Quantum channels of a single qubit act as affine transformations of the Bloch ball:

$$\mathcal{E}(\rho) = \frac{\mathbb{I} + \mathbf{r}' \cdot \boldsymbol{\sigma}}{2}, \quad \mathbf{r}' = M\mathbf{r} + \mathbf{v}. \quad (2.104)$$

where M is a 3×3 matrix and $\mathbf{v}$ is a constant vector.

2.4.2.3 Bit-flip channel

For

$$\mathcal{E}_X(\rho) = (1 - p)\rho + p\sigma_x\rho\sigma_x, \quad (2.105)$$

the Bloch vector transforms as

$$(x, y, z) \mapsto (x, (1 - 2p)y, (1 - 2p)z). \quad (2.106)$$

The Bloch ball contracts toward the x axis.

2.4.2.4 Phase-flip channel and phase damping

The phase-flip channel is

$$\mathcal{E}_Z(\rho) = (1 - p)\rho + p\sigma_z\rho\sigma_z. \quad (2.107)$$

For

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \rho_0 = |\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}, \quad (2.108)$$

one application gives

$$\mathcal{E}_Z(\rho_0) = \begin{pmatrix} |\alpha|^2 & (1 - 2p)\alpha\beta^* \\ (1 - 2p)\alpha^*\beta & |\beta|^2 \end{pmatrix}. \quad (2.109)$$

After n applications the off-diagonal entries acquire a factor $(1 - 2p)^n$. Setting $2p = \Gamma t/n$ and taking $n \rightarrow \infty$ gives

$$\rho(t) = \begin{pmatrix} |\alpha|^2 & e^{-\Gamma t}\alpha\beta^* \\ e^{-\Gamma t}\alpha^*\beta & |\beta|^2 \end{pmatrix}. \quad (2.110)$$

The populations remain fixed while coherence decays.

2.4.2.5 Depolarizing channel

A common parametrization is

$$\mathcal{E}_{\text{dep}}(\rho) = (1 - p)\rho + \frac{p}{3}(\sigma_x\rho\sigma_x + \sigma_y\rho\sigma_y + \sigma_z\rho\sigma_z). \quad (2.111)$$

Using

$$2\mathbb{I} = \rho + \sigma_x\rho\sigma_x + \sigma_y\rho\sigma_y + \sigma_z\rho\sigma_z, \quad (2.112)$$

the Bloch vector transforms as

$$\mathbf{r} \mapsto \left(1 - \frac{4p}{3}\right)\mathbf{r}. \quad (2.113)$$

One Kraus representation is

$$M_0 = \sqrt{1 - p}\mathbb{I}, \quad M_j = \sqrt{\frac{p}{3}}\sigma_j, \quad j = x, y, z. \quad (2.114)$$

2.4.2.6 Amplitude-damping channel

Let $|0\rangle_B$ denote no photon and $|1\rangle_B$ one photon. The joint evolution satisfies

$$U|0\rangle_A|0\rangle_B = |0\rangle_A|0\rangle_B, \quad (2.115)$$

$$U|1\rangle_A|0\rangle_B = \sqrt{1-p}|1\rangle_A|0\rangle_B + \sqrt{p}|0\rangle_A|1\rangle_B. \quad (2.116)$$

The Kraus operators are

$$M_0 = {}_B\langle 0|U|0\rangle_B = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}, \quad M_1 = {}_B\langle 1|U|0\rangle_B = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}. \quad (2.117)$$

Thus

$$\mathcal{E}_{\text{AD}}(\rho_0) = \begin{pmatrix} |\alpha|^2 + p|\beta|^2 & \sqrt{1-p}\alpha\beta^* \\ \sqrt{1-p}\alpha^*\beta & (1-p)|\beta|^2 \end{pmatrix}. \quad (2.118)$$

For $p = \Gamma t/n$ and $n \rightarrow \infty$,

$$\rho(t) = \begin{pmatrix} |\alpha|^2 + (1-e^{-\Gamma t})|\beta|^2 & e^{-\Gamma t/2}\alpha\beta^* \\ e^{-\Gamma t/2}\alpha^*\beta & e^{-\Gamma t}|\beta|^2 \end{pmatrix}. \quad (2.119)$$

The excited-state population decays as $e^{-\Gamma t}$, while coherence decays as $e^{-\Gamma t/2}$.

2.4.3 Limits of the description

A channel acting only on A does not describe every possible reduced evolution. If A and B are initially correlated, assigning a single completely positive map to arbitrary initial states of A may be impossible. The standard channel construction assumes a fixed, initially uncorrelated environment state. The Lindblad equation adds assumptions such as Markovianity and time locality.

2.5 Appendix: Summarizing the postulates of quantum mechanics

Below we summarize the 4 postulates of quantum mechanics using two different languages: (1) in terms of state vectors for a closed quantum system, (2) in terms of density matrices for an open quantum system.

2.5.1 State vector formulation of quantum mechanics for a closed system

Postulate 2.2. *A quantum system is described by a Hilbert space $\mathbb{V}$, and each state of the quantum system is described by a normalized vector $|\psi\rangle \in \mathbb{V}$ in the Hilbert space, up to a global phase factor.*

Postulate 2.3. *Any physical observable corresponds to a Hermitian operator $\hat{O} = \hat{O}^\dagger$ in the Hilbert space, which has a spectral decomposition $\hat{O} = \sum_q q |q\rangle\langle q|$ with real eigenvalues $\{q\}$. A projective measurement of this observable on state $|\psi\rangle$ can yield a measurement outcome of q with probability $Pr(q) = |\langle q|\psi\rangle|^2$, and the initial state $|\psi\rangle$ collapses into state $|q\rangle$ after reading out the measurement outcome q .*

Postulate 2.4. *The time evolution of a closed quantum system is described by the Schrodinger equation with a Hamiltonian $\hat{H}(t)$: $i \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle$. Equivalently $|\psi(t)\rangle = \hat{U}(t, 0) |\psi(0)\rangle$ i.e. the state evolves under a unitary time evolution operator $\hat{U}(t, 0)$.*

Postulate 2.5. *A composite quantum system of two subsystem $\mathbb{V}_1$ and $\mathbb{V}_2$ corresponds to a tensor product of the two Hilbert space $\mathbb{V} = \mathbb{V}_1 \otimes \mathbb{V}_2$. The states and operators follow the tensor product rules similarly.*

2.5.2 Density matrix formulation of quantum mechanics for an open system

Postulate 2.6. *A quantum system is described by a Hilbert space $\mathbb{V}$, and each state of the quantum system corresponds to a density matrix $\hat{\rho} \in L_{\mathbb{V}}$, which is a linear operator satisfying the trace condition $\text{Tr } \hat{\rho} = 1$ and positivity condition $\hat{\rho} \geq 0$.*

Postulate 2.7. *A generalized (POVM) measurement on quantum state ρ is described by a set of measurement operators $\{\hat{M}_q\}$ satisfying the completeness relation $\sum_q \hat{M}_q^\dagger \hat{M}_q = \hat{1}$. The probability for measurement outcome q is given by $Pr(q) = \text{Tr } M_q \rho M_q^\dagger$, and the initial state ρ collapses into (unnormalized) state $M_q \rho M_q^\dagger$ after reading out the measurement outcome q .*

Postulate 2.8. *If the system and environment are not correlated at the beginning, the time evolution of an open quantum system is described by a quantum channel (i.e. a CPTP map). Under further assumptions of Markovianity and time locality, the time evolution is described by a master equation of the Lindblad Form.*

Postulate 2.9. *A composite quantum system of two subsystem $\mathbb{V}_1$ and $\mathbb{V}_2$ corresponds to a tensor product of the two Hilbert space $\mathbb{V} = \mathbb{V}_1 \otimes \mathbb{V}_2$. The states and operators follow the tensor product rules similarly.*

3 A quantum particle in one dimension

3.1 Position, momentum and Schrodinger's wave equation (9/30)

- Refer to Tong 3.2.5, S&N 1.6, 1.7

3.1.1 Position and momentum: from a discrete lattice to the continuum

3.1.1.1 Position basis on a lattice

Consider a one-dimensional lattice with lattice constant a . Its sites are labeled by integers n , and the position eigenstates form the basis of Hilbert space

$$\mathbb{V} = \text{span}\{|n\rangle : n = 0, \pm 1, \pm 2, \dots\} \simeq \mathbb{C}^\infty \quad (3.1)$$

The position operator is

$$\hat{x} = \sum_{n=-\infty}^{\infty} na|n\rangle\langle n|, \quad (3.2)$$

so that

$$\hat{x}|n\rangle = na|n\rangle. \quad (3.3)$$

3.1.1.2 Translation operator

Define the one-site translation operator by

$$\hat{T}_x = \sum_{n=-\infty}^{\infty} |n+1\rangle\langle n|. \quad (3.4)$$

Its adjoint translates in the opposite direction, and

$$\hat{T}_x^\dagger \hat{T}_x = \mathbb{I}, \quad (3.5)$$

so $\hat{T}_x$ is unitary. The position operator and translation operator satisfy

$$[\hat{x}, \hat{T}_x] = a\hat{T}_x. \quad (3.6)$$

The eigenstates of translation are labeled by a dimensionless wave number k :

$$\hat{T}_x|k\rangle = e^{-ik}|k\rangle, \quad 0 \leq k < 2\pi. \quad (3.7)$$

For a periodic lattice with N sites and $|n\rangle \equiv |n+N\rangle$, the change of basis is implemented by a unitary transformation

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{ikn}|n\rangle, \quad k = \frac{2\pi}{N}n_k \pmod{2\pi}, \quad n_k \in \mathbb{Z}. \quad (3.8)$$

known as the discrete Fourier transform (DFT). The inverse transformation is

$$|n\rangle = \frac{1}{\sqrt{N}} \sum_{n_k=0}^{N-1} e^{-ikn}|k\rangle, \quad \langle n_1|n_2\rangle = \delta_{n_1, n_2}. \quad (3.9)$$

We also have the resolution of identity for these two basis sets

$$\sum_{n=0}^{N-1} |n\rangle\langle n| = \sum_{n_k=0}^{N-1} |k\rangle\langle k| = \hat{1} \quad (3.10)$$

3.1.1.3 Momentum as the generator of translation

Because $\hat{T}_x$ is unitary, write

$$\hat{T}_x = e^{-ia\hat{p}/\hbar}. \quad (3.11)$$

The momentum operator is therefore the generator of spatial translations. On a translation eigenstate,

$$\hat{p}|k\rangle = \frac{\hbar k}{a}|k\rangle. \quad (3.12)$$

Using the commutator of $\hat{x}$ with $\hat{T}_x$, or expanding an infinitesimal translation ($a \rightarrow 0^+$), gives the canonical commutation relation

$$\boxed{[\hat{x}, \hat{p}] = i\hbar.} \quad (3.13)$$

3.1.1.4 Continuum and thermodynamic limits

To connect the DFT on a discrete lattice to position and momentum operators in the continuum, we need to simultaneously take the continuum limit

$$a \longrightarrow 0 \quad (3.14)$$

and the thermodynamic limit

$$Na \longrightarrow \infty. \quad (3.15)$$

Introduce continuous variables

$$x = na, \quad p = \frac{\hbar k}{a}, \quad (3.16)$$

with spacings that approach zero in the continuum and thermodynamic limits:

$$\Delta x = a, \quad \Delta p = \frac{2\pi\hbar}{Na}. \quad (3.17)$$

and associated basis vectors:

$$|x\rangle \equiv \frac{1}{\sqrt{a}}|n = \frac{x}{a}\rangle, \quad |p\rangle \equiv \sqrt{\frac{Na}{2\pi\hbar}}|k = \frac{pa}{\hbar}\rangle \quad (3.18)$$

The discrete resolutions of the identity in (3.10) become

$$\int_{-\infty}^{\infty} dp |p\rangle\langle p| = \mathbb{I}, \quad \int_{-\infty}^{\infty} dx |x\rangle\langle x| = \mathbb{I}. \quad (3.19)$$

The continuum states are normalized according to

$$\langle x|x'\rangle = \delta(x - x') = \lim_{\Delta x \rightarrow 0^+} \frac{\delta_{n,n'}}{\Delta x}, \quad \langle p|p'\rangle = \delta(p - p') = \lim_{\Delta p \rightarrow 0^+} \frac{\delta_{p,p'}}{\Delta p}. \quad (3.20)$$

3 A quantum particle in one dimension

Finally, the position-momentum overlap is the plane wave

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}. \quad (3.21)$$

3.1.2 The Schrödinger Wave Equation

3.1.2.1 Fourier transformation between representations

Using the continuum completeness relations,

$$|p\rangle = \int_{-\infty}^{\infty} dx |x\rangle \langle x|p\rangle = \int_{-\infty}^{\infty} dx |x\rangle \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}, \quad (3.22)$$

$$|x\rangle = \int_{-\infty}^{\infty} dp |p\rangle \langle p|x\rangle = \int_{-\infty}^{\infty} dp |p\rangle \frac{e^{-ipx/\hbar}}{\sqrt{2\pi\hbar}}. \quad (3.23)$$

The orthonormality relations follow from

$$\frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dx e^{i(p'-p)x/\hbar} = \delta(p-p'), \quad (3.24)$$

and similarly for $\delta(x-x')$.

3.1.2.2 Dirac delta function

The Dirac delta distribution satisfies

$$\delta(x) = 0 \quad \text{for } x \neq 0, \quad (3.25)$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1, \quad (3.26)$$

$$\delta(x) = \delta(-x), \quad (3.27)$$

$$\int_{-\infty}^{\infty} f(x) \delta(x-x_0) dx = f(x_0). \quad (3.28)$$

A useful Fourier representation is

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk. \quad (3.29)$$

The scaling and derivative identities are

$$\delta(ax) = \frac{1}{|a|} \delta(x), \quad \int f(x) \delta'(x-x_0) dx = -f'(x_0). \quad (3.30)$$

Two common regularized representations of Dirac delta function are

$$\delta(x) = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \frac{\epsilon}{x^2 + \epsilon^2} \quad (3.31)$$

and

$$\delta(x) = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\sqrt{\pi\epsilon}} e^{-x^2/\epsilon}. \quad (3.32)$$

3.1.2.3 Wave functions

For a state $|\psi\rangle$, define its position-space and momentum-space wave functions by

$$\psi(x) = \langle x|\psi\rangle, \quad \tilde{\psi}(p) = \langle p|\psi\rangle. \quad (3.33)$$

They are Fourier-transform pairs:

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{ipx/\hbar} \tilde{\psi}(p), \quad (3.34)$$

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx e^{-ipx/\hbar} \psi(x). \quad (3.35)$$

3.1.2.4 Operators in the position representation

For a function of position,

$$\langle x|V(\hat{x})|x'\rangle = V(x)\delta(x-x'). \quad (3.36)$$

For momentum,

$$\langle x|\hat{p}|x'\rangle = \int dp \langle x|p\rangle p \langle p|x'\rangle \quad (3.37)$$

$$= -i\hbar \frac{\partial}{\partial x} \delta(x-x'). \quad (3.38)$$

Thus, acting on wave functions,

$$\boxed{\hat{p} \longrightarrow -i\hbar \frac{\partial}{\partial x}.} \quad (3.39)$$

More generally,

$$\langle x|U(\hat{p})|x'\rangle = U\left(-i\hbar \frac{\partial}{\partial x}\right) \delta(x-x'). \quad (3.40)$$

3.1.2.5 Time-dependent and stationary Schrödinger equations

The abstract time-dependent Schrödinger equation is

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (3.41)$$

If

$$\hat{H}|n\rangle = E_n|n\rangle, \quad c_n = \langle n|\psi(0)\rangle, \quad (3.42)$$

then

$$|\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |n\rangle. \quad (3.43)$$

The time-independent (or stationary) Schrodinger equation is

$$\hat{H}|\psi\rangle = E|\psi\rangle. \quad (3.44)$$

3.1.2.6 Wave equation for a particle in one dimension

For a particle with no electromagnetic field,

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}). \quad (3.45)$$

3 A quantum particle in one dimension

Taking the position-space matrix element of the stationary equation gives

$$\langle x|\hat{H}|\psi\rangle = \frac{1}{2m}\langle x|\hat{p}^2|\psi\rangle + \langle x|V(\hat{x})|\psi\rangle \quad (3.46)$$

$$= -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) + V(x)\psi(x). \quad (3.47)$$

Therefore,

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x).} \quad (3.48)$$

This is the one-dimensional stationary Schrödinger wave equation.

In momentum space, where $\tilde{\psi}(p) = \langle p|\psi$, the same equation becomes

$$\boxed{\frac{p^2}{2m}\tilde{\psi}(p) + V\left(i\hbar\frac{\partial}{\partial p}\right)\tilde{\psi}(p) = E\tilde{\psi}(p).} \quad (3.49)$$

3.1.3 Summary

- Momentum is the generator of spatial translations.
- The continuum position and momentum bases are delta normalized and connected by a Fourier transform.
- In the position representation, $\hat{p} = -i\hbar\partial/\partial x$.
- In the momentum representation, $\hat{x} = i\hbar\partial/\partial p$.
- Projecting the abstract eigenvalue problem onto a position basis gives the Schrödinger wave equation.

3.2 Solving one-dimensional potentials (10/2)

- Refer to Tong 2.1.1, 2.3.2, 2.4.1; S&N Appendix B on “Elementary Solutions to Schrödinger Wave Equation”

3.2.1 General Properties of stationary Schrödinger equations

The stationary Schrödinger equation is

$$\langle x | \hat{H} | \psi \rangle = E \langle x | \psi \rangle. \quad (3.50)$$

For a particle of mass m in a one-dimensional potential $V(x)$,

$$\boxed{-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x).} \quad (3.51)$$

This is a linear, second-order ordinary differential equation.

Linearity:

If $\psi_1(x)$ and $\psi_2(x)$ are solutions with the same energy E , then any linear combination

$$c_1\psi_1(x) + c_2\psi_2(x), \quad c_1, c_2 \in \mathbb{C}, \quad (3.52)$$

is also a solution with energy E .

Complex conjugation and time reversal symmetry

If $V(x)$ is real and $\psi(x)$ is a solution with energy E , then $\psi^*(x)$ is also a solution with the same energy. For

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad (3.53)$$

this follows from

$$\hat{H}^* = \hat{H}. \quad (3.54)$$

This is the stationary manifestation of time-reversal symmetry for a spinless particle in a real scalar potential.

Real eigenfunctions

Energy eigenfunctions can be chosen to be real. Starting from a complex solution $\psi(x)$, define

$$\psi_1(x) = \psi(x) + \psi^*(x), \quad \psi_2(x) = i(\psi(x) - \psi^*(x)). \quad (3.55)$$

Both ψ_1 and ψ_2 are real solutions with the same energy. A nonzero member of this pair can therefore be used as a real eigenfunction.

Smoothness and matching conditions

If $V(x)$ is smooth, then solutions of the Schrödinger equation are smooth. If $V(x)$ has a finite discontinuity, integration of the equation across the discontinuity shows that both $\psi(x)$ and $\psi'(x)$ are continuous. Singular potentials, such as a delta function, require a modified derivative matching condition.

3.2.2 Examples

3.2.2.1 Free particle

For a free particle,

$$V(x) = 0. \quad (3.56)$$

3 A quantum particle in one dimension

The plane-wave solutions are

$$\psi_k(x) \propto e^{ikx}, \quad E = \frac{\hbar^2 k^2}{2m}. \quad (3.57)$$

These are scattering states and are not square-normalizable on the entire real line. The two waves ψ_{+k} and ψ_{-k} have the same energy. Real eigenfunctions may be chosen as $\cos(kx)$ and $\sin(kx)$.

To properly normalize these plane waves, we have to utilize the Dirac delta function:

$$\langle k_1 | k_2 \rangle = \int dx \psi_{k_1}^*(x) \psi_{k_2}(x) = \delta(k_1 - k_2) \quad (3.58)$$

which means

$$\psi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx} \quad (3.59)$$

Periodic boundary condition:

If we consider a plane wave on a periodic ring of length L , we need to enforce the *periodic boundary condition*

$$\psi_k(x) \equiv \psi_k(x + L) \quad (3.60)$$

which quantizes the wavevector k as

$$k = \frac{2\pi}{L} n_k, \quad n_k \in \mathbb{Z}. \quad (3.61)$$

3.2.2.2 Step potential

Consider

$$V(x) = \begin{cases} 0, & x < 0, \\ U, & x > 0, \end{cases} \quad U > 0. \quad (3.62)$$

Because $V(x) \geq 0$, there are no energy eigenstates with $E < 0$.

Case I: $E > U$

Define

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad k' = \frac{\sqrt{2m(E-U)}}{\hbar}. \quad (3.63)$$

For a wave incident from the left, write

$$\psi(x) = \begin{cases} e^{ikx} + r e^{-ikx}, & x < 0, \\ t e^{ik'x}, & x > 0. \end{cases} \quad (3.64)$$

Continuity of ψ and ψ' at $x = 0$ gives

$$1 + r = t, \quad k(1 - r) = k't. \quad (3.65)$$

Therefore,

$$\boxed{r = \frac{k - k'}{k + k'}, \quad t = \frac{2k}{k + k'} = \frac{2}{1 + k'/k}.} \quad (3.66)$$

The reflection and transmission probabilities are determined by probability currents:

$$R = |r|^2, \quad T = \frac{k'}{k}|t|^2, \quad R + T = 1. \quad (3.67)$$

Case II: $0 < E < U$

Now define

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad \kappa = \frac{\sqrt{2m(U-E)}}{\hbar}, \quad k' = i\kappa. \quad (3.68)$$

The physical solution is

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0, \\ te^{-\kappa x}, & x > 0. \end{cases} \quad (3.69)$$

The growing exponential for $x > 0$ is excluded. The matching formulas are the same as before after substituting $k' = i\kappa$:

$$t = \frac{2k}{k + i\kappa}, \quad r = \frac{k - i\kappa}{k + i\kappa}. \quad (3.70)$$

Thus $|r|^2 = 1$. There is no transmitted probability current, although the wave function penetrates a distance of order κ^{-1} into the classically forbidden region.

3.2.2.3 Delta-function potential

Consider

$$V(x) = D_0 \delta(x). \quad (3.71)$$

The wave function is continuous at the origin. Integrating the stationary Schrödinger equation from $-\epsilon$ to $+\epsilon$ and taking $\epsilon \rightarrow 0^+$ gives

$$\boxed{\psi'(0^+) - \psi'(0^-) = \frac{2mD_0}{\hbar^2}\psi(0).} \quad (3.72)$$

Scattering states: $E > 0$

Let

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad (3.73)$$

and use

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < 0, \\ te^{ikx}, & x > 0. \end{cases} \quad (3.74)$$

Continuity gives

$$1 + r = t. \quad (3.75)$$

The derivative jump condition gives

$$ik(t - 1 + r) = \frac{2mD_0}{\hbar^2}t. \quad (3.76)$$

Define

$$\alpha = \frac{mD_0}{\hbar^2 k}. \quad (3.77)$$

Then

$$\boxed{t = \frac{1}{1 + i\alpha}, \quad r = \frac{-i\alpha}{1 + i\alpha}.} \quad (3.78)$$

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Equivalently,

$$t = \frac{1}{1 - mD_0/(ik\hbar^2)}, \quad r = \frac{mD_0/(ik\hbar^2)}{1 - mD_0/(ik\hbar^2)}. \quad (3.79)$$

The fluxes satisfy

$$|t|^2 + |r|^2 = 1. \quad (3.80)$$

The same scattering result applies to either sign of D_0 as long as $E > 0$.

Bound state for an attractive delta potential

Let $D_0 < 0$ and $E < 0$. Define

$$\kappa = \frac{\sqrt{2m|E|}}{\hbar}. \quad (3.81)$$

Normalizability requires exponential decay at both infinities:

$$\psi(x) = \begin{cases} Ae^{\kappa x}, & x < 0, \\ Be^{-\kappa x}, & x > 0. \end{cases} \quad (3.82)$$

The continuity gives $A = B$ and the derivative jump condition becomes

$$-2\kappa A = \frac{2mD_0}{\hbar^2} A, \quad (3.83)$$

so

$$\kappa = -\frac{mD_0}{\hbar^2} = \frac{m|D_0|}{\hbar^2}. \quad (3.84)$$

Therefore the attractive delta potential has one bound state with energy

$$\boxed{E = -\frac{mD_0^2}{2\hbar^2}}. \quad (3.85)$$

Normalization gives

$$1 = \int_{-\infty}^{\infty} |\psi(x)|^2 dx = \frac{|A|^2}{\kappa}, \quad (3.86)$$

so one may choose

$$\boxed{\psi(x) = \sqrt{\kappa} e^{-\kappa|x|}}. \quad (3.87)$$

3.2.3 Bound and Scattering Spectra

Bound states are normalizable and have discrete energies. For the attractive delta potential, there is a single bound-state energy. In contrast, scattering states are not square-normalizable in the full space and form a continuous energy spectrum. They are instead normalized using delta-function normalization.

3.2.4 Summary

- The stationary Schrödinger equation is a linear, second-order ODE.
- For real potentials, energy eigenfunctions can be chosen real.
- Finite potential discontinuities preserve continuity of both ψ and ψ' .
- A delta potential preserves ψ but produces a finite jump in ψ' .
- Step and delta potentials illustrate reflection, transmission, evanescent waves, and bound-state quantization.

3.2 Solving one-dimensional potentials (10/2)

- Bound states have discrete energies, whereas scattering states form a continuous spectrum.

3.3 Bound states in one dimension (10/14)

- Refer to Tong 2.3, 14.1.4, 7.5

A bound-state wave function is square-normalizable:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty. \quad (3.88)$$

Consequently,

$$\lim_{|x| \rightarrow \infty} \psi(x) = 0. \quad (3.89)$$

For a potential with a well-defined asymptotic value, a bound-state energy lies below the asymptotic continuum threshold:

$$E < \lim_{|x| \rightarrow \infty} V(x). \quad (3.90)$$

Unlike scattering states characterized by a continuous wavevector, bound states have a discrete spectrum.

3.3.1 The Infinite Potential Well

3.3.1.1 Potential and boundary conditions

Consider a box of width L centered at the origin:

$$V(x) = \begin{cases} 0, & |x| < L/2, \\ +\infty, & |x| > L/2. \end{cases} \quad (3.91)$$

The wave function vanishes outside the box and satisfies

$$\psi(-L/2) = \psi(L/2) = 0. \quad (3.92)$$

Inside the box, a general energy eigenfunction is

$$\psi(x) = Ae^{ipx/\hbar} + Be^{-ipx/\hbar}. \quad (3.93)$$

The two boundary conditions give

$$Ae^{ipL/(2\hbar)} + Be^{-ipL/(2\hbar)} = 0, \quad (3.94)$$

$$Ae^{-ipL/(2\hbar)} + Be^{ipL/(2\hbar)} = 0. \quad (3.95)$$

A nontrivial solution requires

$$e^{2ipL/\hbar} = 1, \quad (3.96)$$

so the allowed momenta are quantized.

3.3.1.2 Eigenfunctions and energies

It is convenient to write the normalized eigenfunctions as

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{L}} \sin \left[\frac{n\pi}{L} \left(x + \frac{L}{2} \right) \right], \quad n = 1, 2, 3, \dots,} \quad (3.97)$$

for $|x| < L/2$, with $\psi_n(x) = 0$ outside the well. The energies are

$$E_n = \frac{\pi^2 \hbar^2}{2mL^2} n^2, \quad n = 1, 2, 3, \dots \quad (3.98)$$

The ground state has no interior node, the first excited state has one node, and the state with quantum number n has $n - 1$ interior nodes.

3.3.2 General Theorems for One-Dimensional Bound States

3.3.2.1 Nondegeneracy

The bound-state spectrum of a one-dimensional Schrödinger Hamiltonian is nondegenerate. Suppose ψ_1 and ψ_2 are square-normalizable solutions with the same energy E :

$$-\frac{\hbar^2}{2m}\psi_1'' + V\psi_1 = E\psi_1, \quad (3.99)$$

$$-\frac{\hbar^2}{2m}\psi_2'' + V\psi_2 = E\psi_2. \quad (3.100)$$

Multiply the first equation by ψ_2 , the second by ψ_1 , and subtract. This gives

$$\frac{d}{dx} (\psi_2\psi_1' - \psi_1\psi_2') = 0. \quad (3.101)$$

Thus the Wronskian is constant. Bound-state boundary conditions make this constant zero, so

$$\psi_2\psi_1' = \psi_1\psi_2'. \quad (3.102)$$

Where the wave functions are nonzero,

$$\frac{\psi_1'}{\psi_1} = \frac{\psi_2'}{\psi_2}, \quad (3.103)$$

and hence

$$\psi_1(x) = C\psi_2(x). \quad (3.104)$$

The two solutions therefore represent the same physical state.

3.3.2.2 Node theorems

For one-dimensional bound states:

1. The ground-state wave function has no node at finite x .
2. The m th excited state has exactly m nodes.

Equivalently, if the eigenstates are indexed by $n = 1, 2, 3, \dots$, then the n th state has $n - 1$ nodes.

3.3.3 The Finite Square Well

Consider the symmetric finite well

$$V(x) = \begin{cases} -V_0, & |x| < a, \\ 0, & |x| > a, \end{cases} \quad V_0 > 0. \quad (3.105)$$

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3.3.3.1 Scattering states with $E > 0$

For scattering states with $E > 0$, define

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad k' = \frac{\sqrt{2m(E + V_0)}}{\hbar}. \quad (3.106)$$

A left-incident scattering state has the form

$$\psi(x) = \begin{cases} e^{ikx} + re^{-ikx}, & x < -a, \\ Ae^{ik'x} + Be^{-ik'x}, & |x| < a, \\ te^{ikx}, & x > a. \end{cases} \quad (3.107)$$

Continuity of ψ and ψ' at $x = -a$ gives

$$e^{-ika} + re^{ika} = Ae^{-ik'a} + Be^{ik'a}, \quad (3.108)$$

$$k(e^{-ika} - re^{ika}) = k'(Ae^{-ik'a} - Be^{ik'a}). \quad (3.109)$$

At $x = a$,

$$Ae^{ik'a} + Be^{-ik'a} = te^{ika}, \quad (3.110)$$

$$k'(Ae^{ik'a} - Be^{-ik'a}) = kte^{ika}. \quad (3.111)$$

These matching equations determine the reflection and transmission amplitudes.

Solving the two matching equations at $x = \pm a$ for A and B , we find

$$A = \frac{e^{ik'a}}{2} \left[\left(1 + \frac{k}{k'}\right) e^{-ika} + r \left(1 - \frac{k}{k'}\right) e^{ika} \right] = \frac{e^{-ik'a}}{2} te^{ika} \left(1 + \frac{k}{k'}\right), \quad (3.112)$$

$$B = \frac{e^{-ik'a}}{2} \left[\left(1 - \frac{k}{k'}\right) e^{-ika} + r \left(1 + \frac{k}{k'}\right) e^{ika} \right] = \frac{e^{ik'a}}{2} te^{ika} \left(1 - \frac{k}{k'}\right). \quad (3.113)$$

3.3.3.2 Bound states with $-V_0 < E < 0$

For bound states,

$$-V_0 < E < 0. \quad (3.114)$$

Define

$$\kappa = -ik = \frac{\sqrt{2m|E|}}{\hbar}, \quad k' = \frac{\sqrt{2m(E + V_0)}}{\hbar}. \quad (3.115)$$

These obey

$$\kappa^2 + (k')^2 = \frac{2mV_0}{\hbar^2}. \quad (3.116)$$

Normalizability requires exponential decay outside the well:

$$\psi(x) = \begin{cases} re^{\kappa x}, & x < -a, \\ Ae^{ik'x} + Be^{-ik'x}, & |x| < a, \\ te^{-\kappa x}, & x > a. \end{cases} \quad (3.117)$$

Note that the matching conditions for bound states are exactly the same as for scattering states, except for replacing $k = i\kappa$. Comparing the form of wavefunctions, we see that in bound states

$$|r|, |t| \rightarrow \infty \quad (3.118)$$

and Eqs. (3.112)-(3.113) becomes

$$\frac{r}{t} = e^{2ik'a} \frac{k' - k}{k' + k} = e^{-2ik'a} \frac{k' + k}{k' - k} = \pm 1. \quad (3.119)$$

for bound states.

3.3.3.3 Inversion symmetry

For an even potential,

$$V(x) = V(-x), \quad (3.120)$$

define the parity operator

$$\hat{P} = \int_{-\infty}^{\infty} dx | -x \rangle \langle x|. \quad (3.121)$$

It satisfies

$$\hat{P}^2 = \mathbb{I}, \quad \hat{P}\hat{x}\hat{P}^{-1} = -\hat{x}, \quad \hat{P}\hat{p}\hat{P}^{-1} = -\hat{p}. \quad (3.122)$$

For

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad (3.123)$$

one has

$$[\hat{P}, \hat{H}] = 0. \quad (3.124)$$

Because the one-dimensional bound spectrum is nondegenerate, each bound energy eigenstate is also a parity eigenstate:

$$\hat{P}|\psi\rangle = \frac{r}{t}|\psi\rangle = \pm|\psi\rangle. \quad (3.125)$$

Thus the wave function is either even or odd under the parity operator, depending on the sign of

$$\frac{r}{t} = e^{-2ik'a} \frac{k' + i\kappa}{k' - i\kappa} = \pm 1, \quad \kappa = \sqrt{2mV_0/\hbar^2 - (k')^2} \quad (3.126)$$

3.3.3.4 Even-parity bound states

For an even state, choose

$$\psi(x) = \begin{cases} Ce^{\kappa x}, & x < -a, \\ D \cos(k'x), & |x| < a, \\ Ce^{-\kappa x}, & x > a. \end{cases} \quad (3.127)$$

Matching ψ and ψ' at $x = a$ gives

$$\boxed{k' \tan(k'a) = \kappa.} \quad (3.128)$$

Using $\kappa^2 + (k')^2 = 2mV_0/\hbar^2$,

$$\boxed{k' \tan(k'a) = \sqrt{\frac{2mV_0}{\hbar^2} - (k')^2}.} \quad (3.129)$$

3.3.3.5 Odd-parity bound states

For an odd state, choose

$$\psi(x) = \begin{cases} -Ce^{\kappa x}, & x < -a, \\ D \sin(k'x), & |x| < a, \\ Ce^{-\kappa x}, & x > a. \end{cases} \quad (3.130)$$

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Matching at $x = a$ gives

$$\boxed{-k' \cot(k'a) = \kappa,} \quad (3.131)$$

or

$$\boxed{-k' \cot(k'a) = \sqrt{\frac{2mV_0}{\hbar^2} - (k')^2}.} \quad (3.132)$$

Equivalently,

$$\tan\left(k'a + \frac{\pi}{2}\right) = \frac{\kappa}{k'}. \quad (3.133)$$

3.3.3.6 Graphical solution

The allowed bound-state energies are obtained by solving Eq. (3.126), which can be simplified as

$$k' \tan(k'a) = \sqrt{\frac{r}{t}} \cdot \sqrt{\frac{2mV_0}{\hbar^2} - (k')^2} \quad (3.134)$$

Graphically, as shown in Fig. 3.1, the allowed values of k' are the intersections of the red curve

$$\kappa(k') = \sqrt{\frac{2mV_0}{\hbar^2} - (k')^2} \quad (3.135)$$

with the green even-parity branches $k' \tan(k'a)$ and the blue odd-parity branches $-k' \cot(k'a)$. Intersections with $k' \tan(k'a)$ give even-parity states, while intersections with $-k' \cot(k'a)$ give odd-parity states. The vertical dashed lines indicate the poles at $k'a = \pi/2, \pi, \dots$, and the red curve meets the horizontal axis at

$$k'a = a\sqrt{\frac{2mV_0}{\hbar^2}}. \quad (3.136)$$

Once an allowed k' is found,

$$E = -V_0 + \frac{\hbar^2(k')^2}{2m}. \quad (3.137)$$

Only finitely many intersections occur for a well of finite depth and width, so a finite square well supports finitely many bound states.

Finally, note that in the limit of

$$a \rightarrow 0^+, \quad V_0 \rightarrow +\infty, \quad 2aV_0 \rightarrow |D_0| \quad (3.138)$$

the square potential well discussed here reduces to the delta function potential studied in the last section.

3.3.4 Summary

- Bound states are square-normalizable and decay at spatial infinity.
- One-dimensional bound-state energy levels are nondegenerate.
- The ground state has no nodes, and successive excited states acquire one additional node each.
- For an even potential, nondegenerate bound states have definite parity.
- Finite-well energies follow from separate even- and odd-parity transcendental equations.

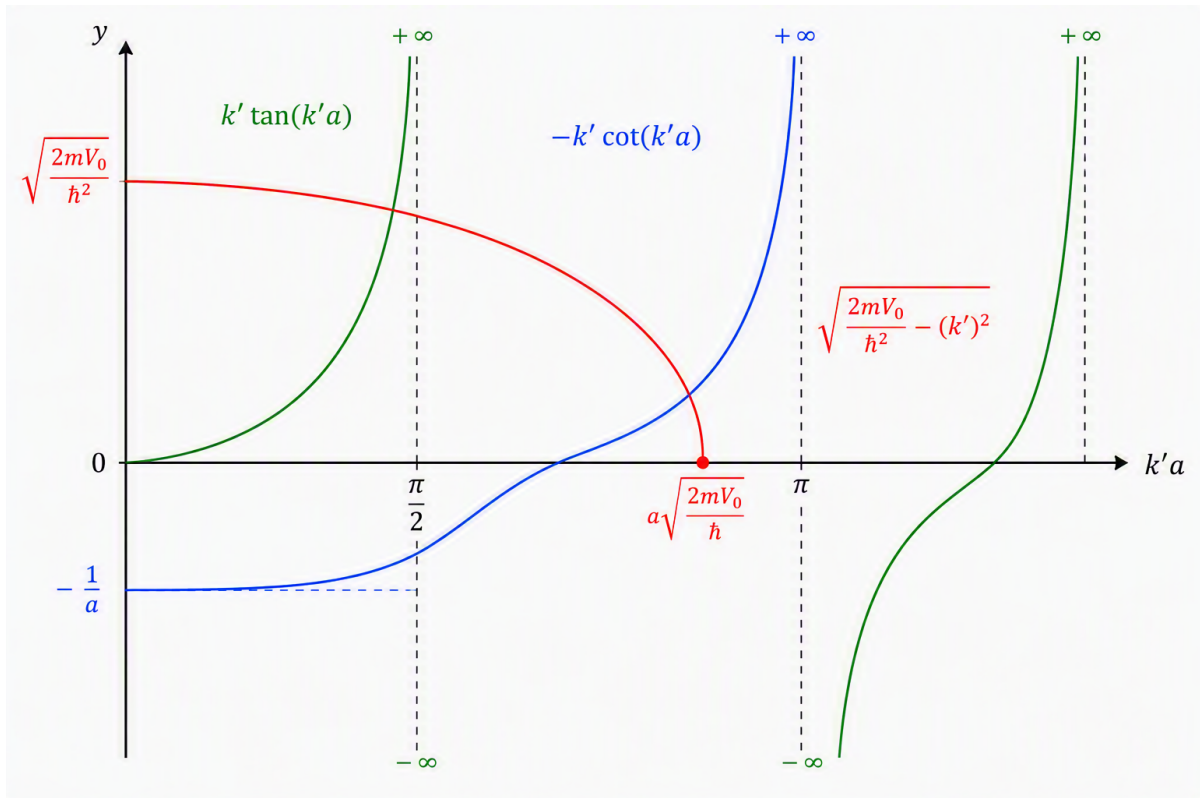


Figure 3.1: Graphical solution of the even- and odd-parity finite-well bound-state equations. Intersections with the red curve determine the allowed bound-state energies.

3.4 Scattering in one dimension (10/21)

- Refer to Tong 14.1.1-14.1.3, S&N 6.1

3.4.1 Scattering by a Delta Potential

We first review the case of a one-dimensional quantum particle scattered by a delta function potential. Consider

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad V(x) = V_0\delta(x). \quad (3.139)$$

If $V_0 > 0$, then $\hat{H} \geq 0$, so a negative-energy bound state is impossible. A bound state with $E < 0$ can occur only for an attractive delta potential.

3.4.1.1 Matching conditions at the singular point

The stationary Schrödinger equation is

$$-\frac{\hbar^2}{2m}\psi''(x) + V_0\delta(x)\psi(x) = E\psi(x). \quad (3.140)$$

The wave function remains continuous,

$$\psi(0^-) = \psi(0^+) = \psi(0), \quad (3.141)$$

but its derivative is discontinuous. Integrating across the origin gives

$$\psi'(0^+) - \psi'(0^-) = \frac{2mV_0}{\hbar^2}\psi(0). \quad (3.142)$$

These two conditions replace the usual continuity of both ψ and ψ' at a finite nonsingular potential discontinuity.

3.4.1.2 Scattering states

Consider a scattering state with $E > 0$, introduce

$$k = \frac{\sqrt{2mE}}{\hbar} > 0. \quad (3.143)$$

A general solution away from the origin is

$$\psi(x) = \begin{cases} \alpha e^{ikx} + \beta e^{-ikx}, & x < 0, \\ \gamma e^{ikx} + \delta e^{-ikx}, & x > 0. \end{cases} \quad (3.144)$$

Here we focus on the **in-state**, i.e. an incidental plane wave from the left

$$\psi_k^+(x) = \begin{cases} e^{ikx} + r e^{-ikx}, & x < 0, \\ t e^{ikx}, & x > 0. \end{cases} \quad (3.145)$$

where r and t correspond to the reflection and transmission amplitude after the incident wave was scattered by the delta function potential.

The continuity and derivative-jump equations are

$$1 + r = t, \quad ikt - ik(1 - r) = \frac{2mV_0}{\hbar^2}t. \quad (3.146)$$

Thus

$$\boxed{t = \frac{1}{1 - \frac{mV_0}{i\hbar^2 k}}, \quad r = \frac{\frac{mV_0}{i\hbar^2 k}}{1 - \frac{mV_0}{i\hbar^2 k}}.} \quad (3.147)$$

Equivalently, with $\eta = mV_0/(\hbar^2 k)$,

$$t = \frac{1}{1 + i\eta}, \quad r = \frac{-i\eta}{1 + i\eta}. \quad (3.148)$$

The reflection and transmission amplitudes obey

$$\boxed{|t|^2 + |r|^2 = 1.} \quad (3.149)$$

3.4.2 General properties of one-Dimensional scattering

3.4.2.1 In-state basis $\{|\psi_k^+\rangle\}$

Let $V(x)$ satisfy

$$\lim_{|x| \rightarrow \infty} V(x) = 0. \quad (3.150)$$

The stationary equation is

$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x). \quad (3.151)$$

For $E > 0$, its asymptotic solutions are plane waves:

$$\psi(x) \sim \alpha e^{ikx} + \beta e^{-ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar}. \quad (3.152)$$

We consider a set of complete orthonormal basis for the scattering state, known as the **in-states** $\{|\psi_k^+\rangle\}$ which physically correspond to the incident plane waves

$$\psi_k^+(x) \sim \begin{cases} e^{ikx} + r e^{-ikx}, & x \rightarrow -\infty, \\ t e^{ikx}, & x \rightarrow +\infty, \end{cases} \quad (3.153)$$

$$\psi_{-k}^+(x) \sim \begin{cases} t' e^{-ikx}, & x \rightarrow -\infty, \\ e^{-ikx} + r' e^{ikx}, & x \rightarrow +\infty. \end{cases} \quad (3.154)$$

If bound states exist, they must also be added to in-states (or out-states) to complete the Hilbert space.

3.4.2.2 Probability flux

For a stationary wave function, define

$$\boxed{j(x) = \frac{i\hbar}{2m} (\psi^* \psi' - \psi (\psi^*)') = \frac{\hbar}{m} \text{Im}(\psi^* \psi').} \quad (3.155)$$

Subtracting the complex-conjugated Schrödinger equation from the original one gives the continuity equation for probability flux in a stationary state (eigenstate):

$$\frac{dj}{dx} = 0. \quad (3.156)$$

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Thus the flux at $x \rightarrow -\infty$ equals the flux at $x \rightarrow +\infty$. Applied to $\psi_{\pm k}^+$, this yields

$$1 = |r|^2 + |t|^2 = |r'|^2 + |t'|^2 \quad (3.157)$$

3.4.2.3 Time-reversal symmetry

For a real scalar potential,

$$\hat{H}^* = \hat{H}. \quad (3.158)$$

Therefore, if $\psi(x)$ has energy E , then $\psi^*(x)$ has the same energy. This produces two independent scattering solutions at each energy $E > 0$.

Because ψ_k^+ and ψ_{-k}^+ have the same energy, form the complex conjugate of a linear combination that cancels one incoming component:

$$(t'\psi_k^+ - r\psi_{-k}^+)^* \propto \psi_{-k}(x) \quad (3.159)$$

This condition, combined with the conservation of probability flux in Eq. (3.157), yields the following relation:

$$t' = t, \quad r' = -\frac{r^*t}{t^*}. \quad (3.160)$$

Together with flux conservation, these relations are equivalent to the unitarity of the two-channel **scattering matrix** (also known as the S -matrix):

$$\hat{S}(k) = \begin{pmatrix} t & r \\ r' & t' \end{pmatrix} = \begin{pmatrix} t & r \\ -\frac{r^*t}{t^*} & t \end{pmatrix}, \quad \hat{S}^\dagger(k)\hat{S}(k) = \hat{1}. \quad (3.161)$$

3.4.2.4 Inversion symmetry in the case of an even potential $V(x) = V(-x)$

The inversion symmetry is implemented by a unitary operator

$$\hat{U} = \int_{-\infty}^{\infty} dx | -x \rangle \langle x |, \quad (3.162)$$

and satisfies

$$\hat{U}\hat{x}\hat{U}^\dagger = -\hat{x}, \quad \hat{U}\hat{p}\hat{U}^\dagger = -\hat{p}. \quad (3.163)$$

If $V(x) = V(-x)$, then $\hat{H} = \hat{U}\hat{H}\hat{U}^\dagger$ i.e. the Hamiltonian $\hat{H}$ preserves the inversion/parity symmetry. Hence, if $\psi(x)$ has energy E , so does $\psi(-x)$.

In the presence of inversion symmetry, the two in-states are related by the inversion operator $\hat{U}|\psi_k^+\rangle = |\psi_{-k}^+\rangle$:

$$\psi_{-k}^+(x) = \psi_k(-x) \quad (3.164)$$

and hence

$$r = r', \quad t = t'. \quad (3.165)$$

3.4.2.5 Out-state basis $\{|\psi_k^-\rangle\}$

The out-states are the time reversal partners of the in-states:

$$\psi_k^-(x) = [\psi_{-k}^+(x)]^*, \quad \psi_{-k}^-(x) = [\psi_k^+(x)]^*. \quad (3.166)$$

They are asymptotically outgoing plane waves. When the potential has no bound states, the out-states $\{|\psi_k^-\rangle\}$ also form a complete set of orthonormal basis of the Hilbert space, similar to

the in-state basis $\{|\psi_k^+\rangle\}$. If bound states exist, they must be added to complete the Hilbert space.

The unitary operator that relates these two basis sets $\{|\psi_k^+\rangle\}$ (in-states) and $\{|\psi_k^-\rangle\}$ (out-states) is exactly the scattering matrix:

$$S_{a,b}(k) = \langle \psi_{ak}^- | \psi_{bk}^+ \rangle, \quad a, b = \pm; \quad (3.167)$$

$$\hat{S}(k) = \begin{pmatrix} \langle \psi_k^- | \psi_k^+ \rangle & \langle \psi_k^- | \psi_{-k}^+ \rangle \\ \langle \psi_{-k}^- | \psi_k^+ \rangle & \langle \psi_{-k}^- | \psi_{-k}^+ \rangle \end{pmatrix} = \begin{pmatrix} t & r \\ -\frac{r^* t}{t^*} & t \end{pmatrix} \quad (3.168)$$

There is a deep reason behind this equation, as we explain below.

3.4.3 *Scattering theory in a nutshell

3.4.3.1 The Lippmann–Schwinger Equation

Consider a generic Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{V} \quad (3.169)$$

where $\hat{H}_0$ can be exactly solved (e.g. $\hat{H}_0 = \frac{\hat{p}^2}{2m}$ as we considered in one-dimensional scatterings), and $\hat{V}$ is a complicated term that makes it difficult to solve the full Hamiltonian $\hat{H}$. If

$$\hat{H}|\psi\rangle = E|\psi\rangle, \quad (3.170)$$

then

$$(E - \hat{H}_0)|\psi\rangle = \hat{V}|\psi\rangle. \quad (3.171)$$

Let $|a\rangle$ be a free eigenstate satisfying

$$\hat{H}_0|a\rangle = E_a|a\rangle. \quad (3.172)$$

The in- and out-states are eigenstates of full Hamiltonian $\hat{H}$ with the same energy E_a

$$(E_a - \hat{H} \pm i\epsilon)|\psi_a^\pm\rangle = 0 \quad (3.173)$$

This can be rewritten as (see HW7)

$$\boxed{|\psi_a^\pm\rangle = |a\rangle + \frac{1}{E_a - \hat{H}_0 \pm i\epsilon} \hat{V}|\psi_a^\pm\rangle = |a\rangle + \hat{G}_0^\pm(E_a) \hat{V}|\psi_a^\pm\rangle, \quad \epsilon \rightarrow 0^+} \quad (3.174)$$

where we defined the free retarded (+) and advanced (−) Green's function by

$$G_0^+(E) = \frac{1}{E - \hat{H}_0 + i\epsilon}, \quad G_0^-(E) = \frac{1}{E - \hat{H}_0 - i\epsilon}. \quad (3.175)$$

Iteration gives the so-called *Born series* and a formal solution to $|\psi_a^\pm\rangle$

$$|\psi_a^\pm\rangle = |a\rangle + G_0^\pm \hat{V}|a\rangle + (G_0^\pm \hat{V})^2|a\rangle + \dots \quad (3.176)$$

$$= \sum_{j=0}^{\infty} (G_0^\pm \hat{V})^j |a\rangle = \frac{1}{1 - G_0^\pm \hat{V}} |a\rangle. \quad (3.177)$$

Equivalently,

$$|\psi_a^\pm\rangle = (E_a - \hat{H} \pm i\epsilon)^{-1} (E_a - \hat{H}_0 \pm i\epsilon) |a\rangle. \quad (3.178)$$

3.4.3.2 Adiabatic switching and the S matrix

Gell-Mann and Low introduced an adiabatically switched interaction

$$\hat{H}(t) = \hat{H}_0 + \hat{V}e^{-\epsilon|t|}, \quad \epsilon \rightarrow 0^+. \quad (3.179)$$

The time evolution operator in the interaction picture is

$$U_I^\epsilon(t, t_0) = e^{i\hat{H}_0 t/\hbar} \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t \hat{H}(t') dt' \right] e^{-i\hat{H}_0 t_0/\hbar}. \quad (3.180)$$

The Gell-Mann–Low construction gives the unnormalized scattering states

$$|\psi_a^+\rangle = \lim_{\substack{t_0 \rightarrow -\infty \\ \epsilon \rightarrow 0^+}} U_I^\epsilon(0, t_0) |a\rangle, \quad (3.181)$$

$$\langle \psi_a^- | = \lim_{\substack{t \rightarrow +\infty \\ \epsilon \rightarrow 0^+}} \langle a | U_I^\epsilon(t, 0). \quad (3.182)$$

Therefore,

$$S_{ab} = \langle \psi_a^- | \psi_b^+ \rangle = \lim_{\substack{t \rightarrow +\infty, t_0 \rightarrow -\infty \\ \epsilon \rightarrow 0^+}} \langle a | U_I^\epsilon(t, t_0) | b \rangle, \quad (3.183)$$

and

$$\hat{S} = \lim_{\epsilon \rightarrow 0^+} U_I^\epsilon(+\infty, -\infty). \quad (3.184)$$

Since time evolution is unitary, the scattering matrix is unitary.

3.4.3.3 The scattering matrix

At fixed energy $E = \hbar^2 k^2 / (2m)$, define

$$S_{ab} = \langle \psi_a^- | \psi_b^+ \rangle. \quad (3.185)$$

In the in- and out-state basis,

$$S(k) = \begin{pmatrix} t & r \\ r' & t \end{pmatrix}. \quad (3.186)$$

The scattering matrix is unitary:

$$S^\dagger S = \mathbb{I}. \quad (3.187)$$

3.4.3.4 Position-space Green function

With plane-wave normalization

$$\langle x | k \rangle = \frac{e^{ikx}}{\sqrt{2\pi}}, \quad \int_{-\infty}^{\infty} dk |k\rangle \langle k| = \mathbb{I}, \quad (3.188)$$

the Lippmann–Schwinger equation in position space is

$$\psi_k^+(x) = \frac{e^{ikx}}{\sqrt{2\pi}} + \int_{-\infty}^{\infty} dx' \langle x | G_0^+(E) | x' \rangle V(x') \psi_k^+(x'). \quad (3.189)$$

The Green-function kernel is

$$\langle x | G_0^\pm(E) | x' \rangle = \int_{-\infty}^{\infty} \frac{dq}{2\pi} \frac{e^{iq(x-x')}}{E - \hbar^2 q^2 / (2m) \pm i\epsilon}. \quad (3.190)$$

The $+i\epsilon$ prescription produces the retarded Green function, while the $-i\epsilon$ prescription produces the advanced Green function.

Application to the delta potential:

For

$$V(x) = V_0\delta(x), \quad (3.191)$$

the integral equation reduces to

$$\psi_k^+(x) = \frac{e^{ikx}}{\sqrt{2\pi}} + V_0 \langle x | G_0^+(E) | 0 \rangle \psi_k^+(0). \quad (3.192)$$

Evaluating the same equation at $x = 0$ gives an algebraic equation for $\psi_k^+(0)$, from which the reflection and transmission amplitudes of the delta potential follow.

3.4.4 Summary

- Delta-potential scattering follows from continuity of ψ and a finite jump in ψ' .
- Probability-flux conservation implies $|r|^2 + |t|^2 = 1$.
- Time reversal gives $t' = t$ and $r' = -r^*t/t^*$ for a general real one-dimensional potential.
- Inversion symmetry further constrains $r' = r$ for scattering by an even potential $V(x) = V(-x)$.
- The Lippmann–Schwinger equation encodes boundary conditions for in- and out-states through the $\pm i\epsilon$ prescriptions.
- The scattering matrix is unitary because the underlying time evolution is unitary.

4 Quantum optics

4.1 Simple harmonic oscillators (10/23)

- Refer to Tong 3.5.1, S&N 1.4, 2.3

4.1.1 Heisenberg's Uncertainty Principle

4.1.1.1 Expectation values and fluctuations

For a normalized pure state $|\psi\rangle$ and a Hermitian operator $\hat{A}$,

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle. \quad (4.1)$$

Define the fluctuation operator

$$\Delta \hat{A} = \hat{A} - \langle \hat{A} \rangle. \quad (4.2)$$

The standard deviation of observable $\hat{A}$ is

$$\Delta A \equiv \sqrt{\langle (\Delta \hat{A})^2 \rangle} = \sqrt{\langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2}. \quad (4.3)$$

4.1.1.2 The uncertainty relation

The **Cauchy-Schwarz inequality** states

$$\langle \psi_1 | \psi_1 \rangle \langle \psi_2 | \psi_2 \rangle \geq |\langle \psi_1 | \psi_2 \rangle|^2, \quad (4.4)$$

with equality if and only if the two vectors are linearly dependent. Choose

$$|\psi_1\rangle = \Delta \hat{A} |\psi\rangle, \quad |\psi_2\rangle = \Delta \hat{B} |\psi\rangle. \quad (4.5)$$

Then

$$(\Delta A)^2 (\Delta B)^2 \geq \left| \langle \Delta \hat{A} \Delta \hat{B} \rangle \right|^2. \quad (4.6)$$

Decompose the product into commutator and anticommutator parts:

$$\Delta \hat{A} \Delta \hat{B} = \frac{1}{2} [\Delta \hat{A}, \Delta \hat{B}] + \frac{1}{2} \{\Delta \hat{A}, \Delta \hat{B}\}. \quad (4.7)$$

For Hermitian $\hat{A}$ and $\hat{B}$, the commutator expectation value is purely imaginary and the anticommutator expectation value is real. Therefore,

$$(\Delta A)^2 (\Delta B)^2 \geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2 + \frac{1}{4} \left| \langle \{\Delta \hat{A}, \Delta \hat{B}\} \rangle \right|^2 \quad (4.8)$$

$$\geq \frac{1}{4} \left| \langle [\hat{A}, \hat{B}] \rangle \right|^2. \quad (4.9)$$

Thus

$$\boxed{\Delta A \Delta B \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right|}. \quad (4.10)$$

4.1.1.3 Conditions for equality

Equality in the Schwarz inequality requires constants a and b , not both zero, such that

$$(a \Delta \hat{A} + b \Delta \hat{B}) |\psi\rangle = 0. \quad (4.11)$$

To saturate the commutator-only uncertainty relation, one also needs

$$\langle \{ \Delta \hat{A}, \Delta \hat{B} \} \rangle = 0. \quad (4.12)$$

The second condition follows from the first when $a^*b + ab^* = 0$ i.e.

$$\arg a = \arg b \pm \frac{\pi}{2} \quad (4.13)$$

4.1.1.4 Spin operators as an example

In this example we consider

$$\hat{A} = \hat{S}_x, \quad \hat{B} = \hat{S}_y. \quad (4.14)$$

Since

$$[\hat{S}_x, \hat{S}_y] = i\hbar \hat{S}_z, \quad \{\hat{S}_x, \hat{S}_y\} = 0 \quad (4.15)$$

for spin one-half, the uncertainty relation is

$$\Delta S_x \cdot \Delta S_y \geq \frac{\hbar}{2} |\langle \hat{S}_z \rangle| \quad (4.16)$$

For an S_x eigenstate, $\Delta S_x = 0$ and consistency requires $\langle \hat{S}_z \rangle = 0$.

4.1.1.5 Minimum-Uncertainty Wave Packets

For

$$\hat{A} = \hat{x}, \quad \hat{B} = \hat{p}, \quad [\hat{x}, \hat{p}] = i\hbar, \quad (4.17)$$

the uncertainty principle becomes

$$\boxed{\Delta x \Delta p \geq \frac{\hbar}{2}.} \quad (4.18)$$

A state saturating the bound obeys

$$\left(\Delta \hat{x} + iR^2 \frac{\Delta \hat{p}}{\hbar} \right) |\psi\rangle = 0. \quad (4.19)$$

Writing

$$x_0 = \langle \hat{x} \rangle, \quad p_0 = \langle \hat{p} \rangle, \quad (4.20)$$

the position-space equation is

$$\left[(x - x_0) + R^2 \left(\frac{d}{dx} - \frac{ip_0}{\hbar} \right) \right] \psi(x) = 0. \quad (4.21)$$

Its normalized solution is the Gaussian packet

$$\boxed{\psi(x) = \frac{1}{\pi^{1/4} \sqrt{R}} \exp \left(\frac{ip_0 x}{\hbar} - \frac{(x - x_0)^2}{2R^2} \right).} \quad (4.22)$$

The position variance is

$$(\Delta x)^2 = \langle (\hat{x} - x_0)^2 \rangle = \frac{R^2}{2}. \quad (4.23)$$

With

$$\langle x|p \rangle = \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}, \quad (4.24)$$

the momentum-space wave function is

$$\tilde{\psi}(p) = \langle p | \psi \rangle = \frac{\sqrt{R}}{\pi^{1/4} \sqrt{\hbar}} \exp \left[-\frac{i(p - p_0)x_0}{\hbar} - \frac{R^2(p - p_0)^2}{2\hbar^2} \right], \quad (4.25)$$

up to an overall phase convention. Hence

$$(\Delta p)^2 = \frac{\hbar^2}{2R^2}, \quad \boxed{\Delta x \Delta p = \frac{\hbar}{2}}. \quad (4.26)$$

4.1.2 The Simple Harmonic Oscillator

The one-dimensional simple harmonic oscillator has the following Hamiltonian

$$\hat{H}_{\text{SHO}} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2. \quad (4.27)$$

Near a stable minimum x_0 , a smooth potential has the expansion

$$V(x) = V(x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \dots. \quad (4.28)$$

Thus the harmonic oscillator is the leading approximation to motion near a stable equilibrium. It is also an exactly solvable model and a building block for systems such as quantized vibrational modes.

4.1.2.1 Ground state is the Gaussian wavepacket

Consider the Gaussian wavepacket $|\psi\rangle$ that saturates the Heisenberg uncertainty relation:

$$\left(\Delta \hat{x} + \frac{iR^2}{\hbar} \Delta \hat{p} \right) |\psi\rangle = 0. \quad (4.29)$$

Then we have

$$\begin{aligned} & \left(\Delta \hat{x} - \frac{iR^2}{\hbar} \Delta \hat{p} \right) \left(\Delta \hat{x} + \frac{iR^2}{\hbar} \Delta \hat{p} \right) |\psi\rangle \\ &= \left[(\Delta \hat{x})^2 + (\Delta \hat{p})^2 \left(\frac{R^2}{\hbar} \right)^2 + \frac{iR^2}{\hbar} [\Delta \hat{x}, \Delta \hat{p}] \right] |\psi\rangle \\ &= \left[(\Delta \hat{x})^2 + (\Delta \hat{p})^2 \left(\frac{R^2}{\hbar} \right)^2 - R^2 \right] |\psi\rangle = 0, \end{aligned} \quad (4.30)$$

where

$$[\Delta \hat{x}, \Delta \hat{p}] = i\hbar. \quad (4.31)$$

Multiplying by $\hbar^2/(2mR^4)$ gives

$$\left[\frac{(\Delta \hat{p})^2}{2m} + \frac{\hbar^2}{2mR^4} (\Delta \hat{x})^2 \right] |\psi\rangle = E |\psi\rangle, \quad (4.32)$$

with the eigen-energy given by

$$E = \frac{\hbar^2}{2mR^2} \quad (4.33)$$

Therefore the parent Hamiltonian for a Gaussian wavepacket is nothing but the simple harmonic

oscillator

$$\hat{H} = \frac{(\Delta\hat{p})^2}{2m} + \frac{m\omega^2}{2}(\Delta\hat{x})^2. \quad (4.34)$$

where the characteristic frequency ω is given by

$$\omega = \frac{\hbar}{mR^2} \quad (4.35)$$

and the ground state energy is

$$E_0 = \frac{\hbar^2}{2mR^2} = \frac{1}{2}\hbar\omega. \quad (4.36)$$

Note the only length scale R is proportional to the width of the wavepacket:

$$R = \sqrt{\frac{\hbar}{m\omega}} = \sqrt{2} \Delta x = \frac{\hbar}{\sqrt{2} \Delta p}. \quad (4.37)$$

4.1.2.2 Ground state wavefunction

In the simple harmonic oscillator Hamiltonian (4.27), we choose $\langle\hat{x}\rangle = \langle\hat{p}\rangle = 0$ and hence $\Delta\hat{x} = \hat{x}$ and $\Delta\hat{p} = \hat{p}$.

In position space,

$$\langle x|\hat{p}|\psi\rangle = -i\hbar \frac{d}{dx}\psi(x). \quad (4.38)$$

The condition (4.29) for ground state $|\psi_0\rangle$ therefore gives

$$\left(\sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}} \frac{d}{dx} \right) \psi_0(x) = 0. \quad (4.39)$$

Hence we arrive at the ground state wavefunction

$$\boxed{\psi_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-m\omega x^2/(2\hbar)} = \frac{1}{\pi^{1/4}\sqrt{R}} e^{-x^2/(2R^2)}.} \quad (4.40)$$

4.1.2.3 Symmetries

The Hamiltonian (4.27) is obviously invariant under the inversion operation $\hat{\pi} \equiv \int dx |x\rangle\langle x|$:

$$\hat{x} \mapsto -\hat{x}, \quad \hat{p} \mapsto -\hat{p}, \quad \hat{H}_{\text{SHO}} \mapsto \hat{H}_{\text{SHO}} \quad (4.41)$$

In addition, the SHO Hamiltonian also preserves another $SO(2) \simeq U(1)$ symmetry:

$$\begin{pmatrix} \hat{x} \\ \hat{p}/m\omega \end{pmatrix} \mapsto \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{p}/m\omega \end{pmatrix} \quad (4.42)$$

where $\theta \in [0, 2\pi)$ is an arbitrary rotation angle.

4.1.3 Equation of motion in the Heisenberg picture

Recall that in the Heisenberg picture, states remain constants while the operators evolve with time:

$$|\psi_H(t)\rangle = |\psi_0\rangle, \quad \hat{O}_H(t) = \hat{U}^\dagger(t,0)\hat{O}\hat{U}(t,0) = e^{i\hat{H}t/\hbar}\hat{O}e^{-i\hat{H}t/\hbar} \quad (4.43)$$

Therefore the operators in the Heisenberg picture obey the following equation of motion:

$$\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t} \hat{O}_H(t) = [\hat{O}_H(t), \hat{H}] = e^{\mathrm{i}\hat{H}t/\hbar} [\hat{O}, \hat{H}] e^{-\mathrm{i}\hat{H}t/\hbar} \quad (4.44)$$

for a time-independent Hamiltonian $\hat{H}$. If we intend to compute the time dependence of expectation values of observable $\hat{O}$, it is more convenient to use the Heisenberg picture.

For example, in the case of a simple harmonic oscillator (4.27), using the canonical commutation relation between position and momentum operators:

$$[\hat{x}, \hat{p}] = \mathrm{i}\hbar \quad (4.45)$$

it is straightforward to show that

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \hat{x}(t) \rangle = \frac{\langle \hat{p}(t) \rangle}{m}, \quad (4.46)$$

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle \hat{p}(t) \rangle = -\langle V'(\hat{x}(t)) \rangle = -m\omega^2 \langle \hat{x}(t) \rangle \quad (4.47)$$

where $V'(x)$ is the derivative of potential $V(x)$ in the Hamiltonian. Solving these two equations of motion produces the harmonic oscillation that is well known in classical mechanics.

4.2 From ladder operators to coherent states (10/28)

- Refer to Tong 5.1, 5.2; S&N 2.3; S&Z 2.1-2.4

4.2.1 Algebraic solution to the simple harmonic oscillator

For the centered oscillator, the minimum-energy Gaussian wavepacket has a width of

$$R = \sqrt{\frac{\hbar}{m\omega}}. \quad (4.48)$$

Label the ground state by $|0\rangle$. The minimum-uncertainty condition is

$$\left(\hat{x} + iR^2\frac{\hat{p}}{\hbar}\right)|0\rangle = 0. \quad (4.49)$$

The commutator of the two conjugate combinations is

$$[\hat{x} + iR^2\hat{p}/\hbar, \hat{x} - iR^2\hat{p}/\hbar] = 2R^2. \quad (4.50)$$

4.2.1.1 Ladder operators

Define

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} + \frac{i}{\sqrt{2\hbar m\omega}}\hat{p}, \quad (4.51)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\hat{x} - \frac{i}{\sqrt{2\hbar m\omega}}\hat{p}. \quad (4.52)$$

They obey

$$\boxed{[\hat{a}, \hat{a}^\dagger] = 1.} \quad (4.53)$$

The inverse relations are

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger), \quad (4.54)$$

$$\hat{p} = -i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a} - \hat{a}^\dagger). \quad (4.55)$$

Substitution into the Hamiltonian gives

$$\boxed{\hat{H}_{\text{SHO}} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).} \quad (4.56)$$

The ground state satisfies

$$\hat{a}|0\rangle = 0, \quad \hat{H}|0\rangle = \frac{1}{2}\hbar\omega|0\rangle. \quad (4.57)$$

The commutators between the Hamiltonian and ladder operators are

$$[\hat{H}, \hat{a}^\dagger] = \hbar\omega\hat{a}^\dagger, \quad [\hat{H}, \hat{a}] = -\hbar\omega\hat{a}. \quad (4.58)$$

If

$$\hat{H}|\psi\rangle = E|\psi\rangle, \quad (4.59)$$

then

$$\hat{H}\hat{a}^\dagger|\psi\rangle = (E + \hbar\omega)\hat{a}^\dagger|\psi\rangle, \quad (4.60)$$

$$\hat{H}\hat{a}|\psi\rangle = (E - \hbar\omega)\hat{a}|\psi\rangle. \quad (4.61)$$

Thus $\hat{a}^\dagger$ is the creation or raising operator, because it raises the energy by $\hbar\omega$. Meanwhile $\hat{a}$ is the annihilation or lowering operator, since it lowers the energy by $\hbar\omega$.

4.2.1.2 Occupation number basis

The normalized energy eigenstates are

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \quad n = 0, 1, 2, \dots \quad (4.62)$$

They satisfy

$$\hat{H}|n\rangle = \left(n + \frac{1}{2}\right)\hbar\omega|n\rangle. \quad (4.63)$$

The ladder actions are

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle, \quad (4.64)$$

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle. \quad (4.65)$$

Equivalently,

$$\hat{a}^\dagger = \sum_{n=0}^{\infty} \sqrt{n+1}|n+1\rangle\langle n|, \quad (4.66)$$

$$\hat{a} = \sum_{n=0}^{\infty} \sqrt{n+1}|n\rangle\langle n+1|. \quad (4.67)$$

4.2.1.3 Oscillator Wave Functions

Ground state:

In position space,

$$\langle x|\hat{p}|\psi\rangle = -i\hbar\frac{d}{dx}\psi(x). \quad (4.68)$$

The condition $\hat{a}|0\rangle = 0$ gives

$$\left(\sqrt{\frac{m\omega}{2\hbar}}x + \sqrt{\frac{\hbar}{2m\omega}}\frac{d}{dx}\right)\psi_0(x) = 0. \quad (4.69)$$

Hence

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/(2\hbar)} = \frac{1}{\pi^{1/4}\sqrt{R}} e^{-x^2/(2R^2)}. \quad (4.70)$$

First excited state:

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Using $|1\rangle = \hat{a}^\dagger|0\rangle$,

$$\psi_1(x) = \langle x|\hat{a}^\dagger|0\rangle \quad (4.71)$$

$$= \left(\sqrt{\frac{m\omega}{2\hbar}} x - \sqrt{\frac{\hbar}{2m\omega}} \frac{d}{dx} \right) \psi_0(x) \quad (4.72)$$

$$= \sqrt{\frac{2m\omega}{\hbar}} x \psi_0(x). \quad (4.73)$$

Thus

$$\psi_1(x) \propto x e^{-x^2/(2R^2)}. \quad (4.74)$$

Second excited state:

The second excited state is

$$|2\rangle = \frac{(\hat{a}^\dagger)^2}{\sqrt{2!}}|0\rangle = \frac{\hat{a}^\dagger}{\sqrt{2}}|1\rangle. \quad (4.75)$$

Applying the creation operator again gives

$$\boxed{\psi_2(x) = \frac{1}{\sqrt{2}} \left(\frac{2x^2}{R^2} - 1 \right) \psi_0(x).} \quad (4.76)$$

4.2.2 Coherent States

A coherent state is an eigenstate of the annihilation operator:

$$\boxed{\hat{a}|z\rangle = z|z\rangle}, \quad z \in \mathbb{C}. \quad (4.77)$$

Using

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger), \quad \hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}^\dagger - \hat{a}), \quad (4.78)$$

one obtains

$$\langle z|\hat{x}|z\rangle = \sqrt{\frac{2\hbar}{m\omega}} \operatorname{Re} z, \quad (4.79)$$

$$\langle z|\hat{p}|z\rangle = \sqrt{2m\hbar\omega} \operatorname{Im} z. \quad (4.80)$$

4.2.2.1 Time evolution

$$|z(t)\rangle = e^{-i\hat{H}t/\hbar}|z\rangle. \quad (4.81)$$

Since

$$\hat{a}|z(t)\rangle = e^{-i\omega t} z|z(t)\rangle, \quad (4.82)$$

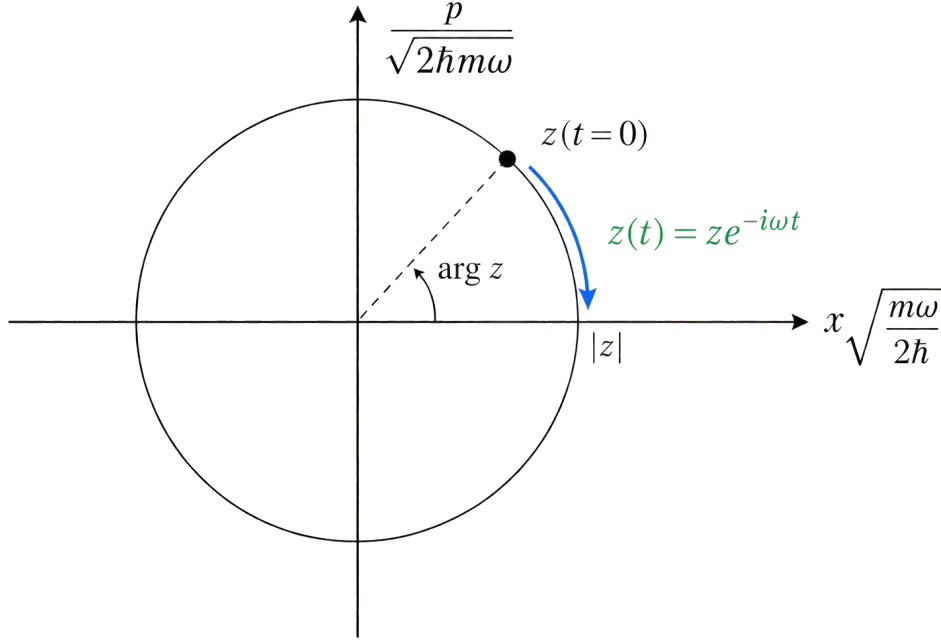
we have

$$\boxed{z(t) = z e^{-i\omega t}}. \quad (4.83)$$

Thus $\langle \hat{x}(t) \rangle$ and $\langle \hat{p}(t) \rangle$ follow the classical oscillator trajectory (see Fig. 4.1):

$$\langle \hat{x}(t) \rangle = \sqrt{\frac{2\hbar}{m\omega}} \operatorname{Re}(z e^{-i\omega t}), \quad (4.84)$$

$$\langle \hat{p}(t) \rangle = \sqrt{2m\hbar\omega} \operatorname{Im}(z e^{-i\omega t}) \quad (4.85)$$


 Figure 4.1: Phase-space evolution of a coherent state, $z(t) = ze^{-i\omega t}$.

4.2.2.2 Number-state expansion

Write

$$|z\rangle = \sum_{n=0}^{\infty} C_n |n\rangle. \quad (4.86)$$

The eigenvalue equation gives

$$C_{n+1} = \frac{z}{\sqrt{n+1}} C_n \implies C_n = \frac{z^n}{\sqrt{n!}} C_0. \quad (4.87)$$

The normalized coherent state is

$$|z\rangle = e^{-|z|^2/2} \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n!}} |n\rangle = e^{-|z|^2/2} e^{z\hat{a}^\dagger} |0\rangle. \quad (4.88)$$

Equivalently,

$$|z\rangle = \hat{D}(z) |0\rangle \quad (4.89)$$

where $D(z)$ is known as the displacement operator:

$$\hat{D}(z) = e^{z\hat{a}^\dagger - z^*\hat{a}}. \quad (4.90)$$

It shifts the center of the Gaussian wavepacket from $\langle \hat{x} \rangle = \langle \hat{p} \rangle = 0$ to

$$z = \sqrt{\frac{m\omega}{2\hbar}} x_0 + \frac{i}{\sqrt{2m\hbar\omega}} p_0 \quad (4.91)$$

4.2.2.3 Overlap and completeness

$$\langle z|z'\rangle = \exp \left[-\frac{|z|^2}{2} - \frac{|z'|^2}{2} + z^* z' \right], \quad (4.92)$$

$$\int_{\mathbb{C}} \frac{d^2 z}{\pi} |z\rangle \langle z| = \mathbb{I}. \quad (4.93)$$

Coherent states form a nonorthogonal overcomplete basis.

4.2.2.4 Gaussian wavepacket as its wavefunction

Let

$$R = \sqrt{\frac{\hbar}{m\omega}}, \quad x_0 = \sqrt{\frac{2\hbar}{m\omega}} \operatorname{Re} z, \quad p_0 = \sqrt{2m\hbar\omega} \operatorname{Im} z. \quad (4.94)$$

Then

$$\left[(\hat{x} - x_0) + \frac{iR^2}{\hbar} (\hat{p} - p_0) \right] |z\rangle = 0. \quad (4.95)$$

Hence we obtain the wavefunction of coherent states as a Gaussian wavepacket:

$$\langle x|z\rangle \propto \exp \left(\frac{ip_0 x}{\hbar} - \frac{(x - x_0)^2}{2R^2} \right), \quad (4.96)$$

$$\langle p|z\rangle \propto \exp \left(-\frac{ix_0 p}{\hbar} - \frac{R^2(p - p_0)^2}{2\hbar^2} \right). \quad (4.97)$$

Since all coherent states saturate the Heisenberg uncertainty relation, among all quantum states, they are closest to the classical trajectory $x(t), p(t)$ of a simple harmonic oscillator.

4.3 Quantization of electromagnetic fields (10/30)

- Refer to S&N 7.6, Tong 5.3, 13.4.1, S&Z 1.1

4.3.1 Classical Electromagnetic Field

Let

$$x^a = (ct, \mathbf{r}), \quad A^a = (\phi/c, \mathbf{A}), \quad (4.98)$$

and

$$F_{ab} = \frac{\partial A_b}{\partial x^a} - \frac{\partial A_a}{\partial x^b}. \quad (4.99)$$

The electromagnetic Lagrangian density is

$$\mathcal{L} = -\frac{1}{4\mu} F^{ab} F_{ab} = \frac{1}{2} \left(\epsilon \mathbf{E}^2 - \frac{\mathbf{B}^2}{\mu} \right), \quad c = \frac{1}{\sqrt{\mu\epsilon}}. \quad (4.100)$$

The electric and magnetic fields are

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t} - \nabla \phi, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (4.101)$$

They are invariant under the following gauge transformations

$$\mathbf{A} \mapsto \mathbf{A} + \nabla \Theta, \quad \phi \mapsto \phi - \frac{\partial \Theta}{\partial t}. \quad (4.102)$$

Below we choose the Weyl and Coulomb gauges:

$$\phi = 0, \quad \nabla \cdot \mathbf{A} = 0. \quad (4.103)$$

4.3.2 One-dimensional optical cavity

In a cavity which is confined along x, z direction but has length L along y direction, we consider one linearly polarized mode propagating along y ,

$$\mathbf{A} = (A(y, t), 0, 0), \quad \mathbf{E} = (E_x, 0, 0), \quad \mathbf{B} = (0, 0, B_z), \quad (4.104)$$

with

$$E_x = -\frac{\partial A}{\partial t}, \quad B_z = -\frac{\partial A}{\partial y}. \quad (4.105)$$

The cavity Lagrangian is

$$L = \int_0^L dy \left[\frac{\epsilon}{2} \left(\frac{\partial A}{\partial t} \right)^2 - \frac{1}{2\mu} \left(\frac{\partial A}{\partial y} \right)^2 \right]. \quad (4.106)$$

4.3.2.1 Standing wave boundary conditions

Impose perfectly reflecting standing-wave boundary conditions

$$\boxed{A(0, t) = A(L, t) = 0.} \quad (4.107)$$

The normalized spatial modes are

$$u_n(y) = \sqrt{\frac{2}{L}} \sin(k_n y), \quad k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots, \quad (4.108)$$

4 Quantum optics

and satisfy

$$\int_0^L dy u_n(y) u_m(y) = \delta_{nm}. \quad (4.109)$$

Expand the field as

$$A(y, t) = \sum_{n=1}^{\infty} q_n(t) u_n(y) = \sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} q_n(t) \sin(k_n y). \quad (4.110)$$

Then

$$E_x(y, t) = -\sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} \dot{q}_n(t) \sin(k_n y), \quad (4.111)$$

$$B_z(y, t) = -\sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} k_n q_n(t) \cos(k_n y). \quad (4.112)$$

4.3.2.2 Mode Lagrangian

Using

$$\int_0^L dy \sin(k_n y) \sin(k_m y) = \frac{L}{2} \delta_{nm}, \quad (4.113)$$

$$\int_0^L dy \cos(k_n y) \cos(k_m y) = \frac{L}{2} \delta_{nm}, \quad (4.114)$$

one obtains the following Lagrangian

$$L = \sum_{n=1}^{\infty} \left(\frac{\epsilon}{2} \dot{q}_n^2 - \frac{k_n^2}{2\mu} q_n^2 \right). \quad (4.115)$$

Thus there is one real harmonic oscillator for each standing-wave mode.

4.3.3 Canonical Quantization

The momentum conjugate to q_n is

$$p_n = \frac{\partial L}{\partial \dot{q}_n} = \epsilon \dot{q}_n. \quad (4.116)$$

Canonical quantization gives

$$[\hat{q}_n, \hat{p}_m] = i\hbar \delta_{nm}, \quad [\hat{q}_n, \hat{q}_m] = 0, \quad [\hat{p}_n, \hat{p}_m] = 0. \quad (4.117)$$

The Hamiltonian is

$$\hat{H} = \sum_{n=1}^{\infty} \left(\hat{p}_n \dot{\hat{q}}_n - L_n \right) \quad (4.118)$$

$$= \sum_{n=1}^{\infty} \left(\frac{\hat{p}_n^2}{2\epsilon} + \frac{k_n^2}{2\mu} \hat{q}_n^2 \right). \quad (4.119)$$

Identifying each term with

$$\frac{\hat{p}_n^2}{2M} + \frac{1}{2}M\omega_n^2\hat{q}_n^2 \quad (4.120)$$

gives

$$M = \epsilon, \quad \omega_n^2 = \frac{k_n^2}{\mu\epsilon} = c^2 k_n^2. \quad (4.121)$$

Therefore,

$$\boxed{\omega_n = ck_n = \frac{n\pi c}{L}.} \quad (4.122)$$

4.3.3.1 Creation and annihilation operators

Define

$$\hat{a}_n = \sqrt{\frac{\epsilon\omega_n}{2\hbar}}\hat{q}_n + \frac{i}{\sqrt{2\hbar\epsilon\omega_n}}\hat{p}_n, \quad (4.123)$$

$$\hat{a}_n^\dagger = \sqrt{\frac{\epsilon\omega_n}{2\hbar}}\hat{q}_n - \frac{i}{\sqrt{2\hbar\epsilon\omega_n}}\hat{p}_n. \quad (4.124)$$

Then

$$[\hat{a}_n, \hat{a}_m^\dagger] = \delta_{nm}. \quad (4.125)$$

The inverse relations are

$$\hat{q}_n = \sqrt{\frac{\hbar}{2\epsilon\omega_n}}(\hat{a}_n + \hat{a}_n^\dagger), \quad (4.126)$$

$$\hat{p}_n = -i\sqrt{\frac{\hbar\epsilon\omega_n}{2}}(\hat{a}_n - \hat{a}_n^\dagger). \quad (4.127)$$

Thus

$$\boxed{\hat{H} = \sum_{n=1}^{\infty} \hbar\omega_n \left(\hat{a}_n^\dagger \hat{a}_n + \frac{1}{2} \right).} \quad (4.128)$$

The total zero-point energy is

$$E_0 = \frac{1}{2} \sum_{n=1}^{\infty} \hbar\omega_n. \quad (4.129)$$

4.3.3.2 Photon number states

For one mode,

$$\hat{N}_n = \hat{a}_n^\dagger \hat{a}_n, \quad \hat{N}_n |N_n\rangle = N_n |N_n\rangle. \quad (4.130)$$

Each excitation carries energy $\hbar\omega_n$, and

$$E_n(N_n) = \hbar\omega_n \left(N_n + \frac{1}{2} \right). \quad (4.131)$$

4.3.4 Quantized Standing-Wave Fields

4.3.4.1 Vector potential

In the Schrödinger picture,

$$\hat{A}(y) = \sum_{n=1}^{\infty} \sqrt{\frac{\hbar}{\epsilon L \omega_n}} \sin(k_n y) (\hat{a}_n + \hat{a}_n^\dagger). \quad (4.132)$$

In the Heisenberg picture,

$$\hat{A}(y, t) = \sum_{n=1}^{\infty} \sqrt{\frac{\hbar}{\epsilon L \omega_n}} \sin(k_n y) \left(\hat{a}_n e^{-i\omega_n t} + \hat{a}_n^\dagger e^{i\omega_n t} \right). \quad (4.133)$$

The boundary conditions are automatically satisfied because

$$\sin(k_n 0) = \sin(k_n L) = 0. \quad (4.134)$$

4.3.4.2 Electric and magnetic fields

Using $\hat{E}_x = -\partial_t \hat{A}$,

$$\hat{E}_x(y, t) = i \sum_{n=1}^{\infty} \sqrt{\frac{\hbar \omega_n}{\epsilon L}} \sin(k_n y) \left(\hat{a}_n e^{-i\omega_n t} - \hat{a}_n^\dagger e^{i\omega_n t} \right). \quad (4.135)$$

Using $\hat{B}_z = -\partial_y \hat{A}$,

$$\hat{B}_z(y, t) = - \sum_{n=1}^{\infty} k_n \sqrt{\frac{\hbar}{\epsilon L \omega_n}} \cos(k_n y) \left(\hat{a}_n e^{-i\omega_n t} + \hat{a}_n^\dagger e^{i\omega_n t} \right). \quad (4.136)$$

4.4 *Squeezed states of light (11/4)

- Refer to S&Z 2.5-2.7

4.4.1 Squeezed states

4.4.1.1 Field quadratures

For a single mode, define the field quadratures:

$$\hat{X} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}}, \quad \hat{Y} = \frac{i(\hat{a}^\dagger - \hat{a})}{\sqrt{2}}. \quad (4.137)$$

Then

$$[\hat{X}, \hat{Y}] = i, \quad \Delta X \Delta Y \geq \frac{1}{2}. \quad (4.138)$$

For coherent states,

$$\Delta X = \Delta Y = \frac{1}{\sqrt{2}}. \quad (4.139)$$

4.4.1.2 Squeezing

A minimum-uncertainty squeezed state satisfies

$$(\Delta \hat{X} + iR^2 \Delta \hat{Y}) |\psi\rangle = 0. \quad (4.140)$$

Equivalently,

$$\left[(1 - R^2) \hat{a}^\dagger + (1 + R^2) \hat{a} - (X_0 + iR^2 Y_0) \right] |\psi\rangle = 0. \quad (4.141)$$

Define the squeeze parameter by

$$\frac{1 - R^2}{1 + R^2} = \tanh r, \quad R = e^{-r}. \quad (4.142)$$

Then

$$\Delta X = \frac{e^{-r}}{\sqrt{2}}, \quad \Delta Y = \frac{e^r}{\sqrt{2}}, \quad \Delta X \Delta Y = \frac{1}{2}. \quad (4.143)$$

4.4.1.3 Electric-field noise

For phase

$$\Phi = \omega t - \mathbf{k} \cdot \mathbf{r}, \quad (4.144)$$

the electric field of a propagating (non-standing) wave

$$\hat{E}(\mathbf{r}, t) = \sqrt{\frac{\hbar \omega}{\epsilon L}} \left(\hat{a} e^{-i(\omega t - \mathbf{k} \cdot \mathbf{r})} - \hat{a}^\dagger e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})} \right). \quad (4.145)$$

can be rewritten as a superposition of quadratures:

$$\hat{E}(\mathbf{r}, t) = \sqrt{\frac{2\hbar \omega}{\epsilon V}} \left(\hat{X} \cos \Phi + \hat{Y} \sin \Phi \right). \quad (4.146)$$

For a squeezed state with vanishing quadrature covariance,

$$(\Delta E)^2 = \frac{\hbar \omega}{\epsilon V} \left(e^{-2r} \cos^2 \Phi + e^{2r} \sin^2 \Phi \right). \quad (4.147)$$

For $r > 0$,

$$\Delta E_{\min} = \Delta X = \frac{e^{-r}}{\sqrt{2}}, \quad \Phi = 0 \pmod{\pi}, \quad (4.148)$$

$$\Delta E_{\max} = \Delta Y = \frac{e^r}{\sqrt{2}}, \quad \Phi = \frac{\pi}{2} \pmod{\pi}, \quad (4.149)$$

$$\Delta E_{\min} \Delta E_{\max} = \Delta X \Delta Y = \frac{1}{2}. \quad (4.150)$$

4.4.2 Generation of coherent and squeezed states in quantum optics

4.4.2.1 Generating coherent states

With a classical current,

$$\hat{H} = \hat{H}_{\text{EM}} - \int \mathbf{j}(\mathbf{r}, t) \cdot \hat{\mathbf{A}}(\mathbf{r}) d^3\mathbf{r}. \quad (4.151)$$

The evolved vacuum is a coherent state:

$$e^{-i\hat{H}t/\hbar}|0\rangle = \exp\left[\sum_k (\alpha_k \hat{a}_k^\dagger - \alpha_k^* \hat{a}_k)\right] |0\rangle, \quad (4.152)$$

where

$$\alpha_k = - \int d^3\mathbf{r} \int_0^t dt' e^{i(\omega t' - \mathbf{k} \cdot \mathbf{r})} \mathbf{j}(\mathbf{r}, t') \cdot \mathbf{e}_k \sqrt{\frac{\hbar}{\omega \epsilon V}}. \quad (4.153)$$

The displacement operator is

$$\boxed{\hat{D}(\alpha) = e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}}}. \quad (4.154)$$

4.4.2.2 Generating squeezed states

Degenerate parametric amplification, achieved by placing a nonlinear crystal nonlinear inside an optical cavity, produces the effective Hamiltonian

$$\hat{H}_{\text{dpa}} = \hbar \chi \beta_0 \left[(\hat{a}^\dagger)^2 e^{-i\phi} + \hat{a}^2 e^{i\phi} \right]. \quad (4.155)$$

Define the squeeze operator:

$$\boxed{\hat{S}(\xi) = \exp\left[\frac{1}{2} \left(\xi^* \hat{a}^2 - \xi (\hat{a}^\dagger)^2 \right)\right]}, \quad \xi = r e^{i\theta}. \quad (4.156)$$

It satisfies

$$\hat{S}^\dagger(\xi)(\hat{X} + i\hat{Y})\hat{S}(\xi) = e^{-r}\hat{X} + ie^r\hat{Y} \quad (4.157)$$

for a suitable squeezing phase. A squeezed coherent state is

$$\boxed{|\xi, z\rangle = \hat{S}(\xi)\hat{D}(z)|0\rangle \sim \hat{S}(\xi)|z\rangle}. \quad (4.158)$$

Its quadrature variances are

$$\langle \xi, z | (\Delta \hat{X})^2 | \xi, z \rangle = \frac{e^{-2r}}{2}, \quad \langle \xi, z | (\Delta \hat{Y})^2 | \xi, z \rangle = \frac{e^{2r}}{2}. \quad (4.159)$$

4.4.3 Application: squeezed light in LIGO

4.4.3.1 Gravitational-wave measurement

LIGO is a Michelson interferometer with long optical cavities in its two perpendicular arms. A passing gravitational wave changes the arm lengths in opposite directions. If the unperturbed arm length is L , define the gravitational-wave strain by

$$h(t) = \frac{\Delta L_x(t) - \Delta L_y(t)}{L}. \quad (4.160)$$

The differential change in arm length produces a phase shift in the light returning from the two arms. The output photodetector measures this phase shift.

Ideally, the interferometer is operated near a dark fringe, so little classical carrier light exits through the antisymmetric port. Quantum vacuum fluctuations can nevertheless enter through this port and interfere with the carrier field. These fluctuations contribute quantum noise to the measured signal.

4.4.3.2 Two sources of quantum noise

Introduce the amplitude and phase quadratures

$$\hat{X} = \frac{\hat{a} + \hat{a}^\dagger}{\sqrt{2}}, \quad \hat{Y} = \frac{i(\hat{a}^\dagger - \hat{a})}{\sqrt{2}}, \quad (4.161)$$

with

$$[\hat{X}, \hat{Y}] = i, \quad \Delta X \Delta Y \geq \frac{1}{2}. \quad (4.162)$$

The two quadratures affect LIGO in different ways:

1. The phase quadrature $\hat{Y}$ produces uncertainty in the optical phase measured at the output. This is called shot noise.
2. The amplitude quadrature $\hat{X}$ produces fluctuations in the number of photons incident on the mirrors. The corresponding fluctuating radiation pressure moves the mirrors. This is called quantum radiation-pressure noise.

The approximate frequency dependence of the quadrature variance is

$$S_h^{\text{shot}}(\Omega) \sim \Delta Y^2 \propto \frac{1}{P}, \quad S_h^{\text{rad}}(\Omega) \sim \Delta X^2 \propto \frac{P}{\Omega^4}, \quad (4.163)$$

where P is the circulating optical power and Ω is the gravitational-wave angular frequency. Shot noise is most important at high frequencies, while radiation-pressure noise becomes important at low frequencies.

Increasing the laser power reduces shot noise but increases radiation-pressure noise. This tradeoff leads to the standard quantum limit.

4.4.3.3 Injection of squeezed vacuum

For ordinary vacuum,

$$(\Delta X)^2 = (\Delta Y)^2 = \frac{1}{2}. \quad (4.164)$$

A squeezed vacuum state redistributes these uncertainties. For phase squeezing,

$$(\Delta X)^2 = \frac{1}{2}e^{2r}, \quad (\Delta Y)^2 = \frac{1}{2}e^{-2r}, \quad (4.165)$$

where $r > 0$ is the squeeze parameter. Therefore,

$$\Delta X \Delta Y = \frac{1}{2}. \quad (4.166)$$

Injecting this squeezed vacuum into the antisymmetric port reduces the phase uncertainty:

$$\Delta Y_{\text{sq}} = e^{-r} \Delta Y_{\text{vac}}. \quad (4.167)$$

The shot-noise contribution is consequently reduced by

$$S_{h,\text{sq}}^{\text{shot}} = e^{-2r} S_{h,\text{vac}}^{\text{shot}}. \quad (4.168)$$

The uncertainty principle is not violated. The reduction of phase noise is accompanied by an increase in amplitude noise:

$$\Delta X_{\text{sq}} = e^r \Delta X_{\text{vac}}, \quad S_{h,\text{sq}}^{\text{rad}} = e^{2r} S_{h,\text{vac}}^{\text{rad}}. \quad (4.169)$$

The increased amplitude noise can increase radiation-pressure noise at low frequencies.

4.4.3.4 Frequency-dependent squeezing

A fixed squeezing angle cannot reduce both forms of quantum noise over the entire detection band. LIGO therefore uses frequency-dependent squeezing.

Let the quadrature vector be

$$\hat{\mathbf{q}} = \begin{pmatrix} \hat{X} \\ \hat{Y} \end{pmatrix}. \quad (4.170)$$

A rotation of the squeezing ellipse through an angle θ is represented by

$$\hat{\mathbf{q}}_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \hat{\mathbf{q}}. \quad (4.171)$$

Before the squeezed vacuum enters the main interferometer, it is reflected from an auxiliary optical filter cavity. The cavity produces a frequency-dependent quadrature rotation,

$$\theta = \theta(\Omega). \quad (4.172)$$

The squeezing ellipse is therefore oriented differently for different gravitational-wave frequencies.

At high frequencies, the squeezed quadrature is aligned to reduce phase noise:

$$(\Delta Y)_{\text{high frequency}}^2 = \frac{1}{2} e^{-2r}. \quad (4.173)$$

This reduces shot noise.

At low frequencies, the filter cavity rotates the ellipse so that the amplitude quadrature is squeezed:

$$(\Delta X)_{\text{low frequency}}^2 = \frac{1}{2} e^{-2r}. \quad (4.174)$$

This reduces radiation-pressure fluctuations acting on the mirrors.

Schematically,

low frequency: amplitude squeezing	$\implies$ less radiation-pressure noise,	(4.175)
high frequency: phase squeezing	$\implies$ less shot noise.	

4.4.3.5 Optical loss

Optical loss replaces part of the squeezed state with ordinary vacuum. If η is the total optical efficiency, the observed squeezed variance is approximately

$$(\Delta Q)_{obs}^2 = \eta \frac{e^{-2r}}{2} + (1 - \eta) \frac{1}{2}, \quad (4.176)$$

where $\hat{Q}$ denotes the intended squeezed quadrature.

Even if the generated squeezing is very strong, the measured noise reduction is limited by optical loss and fluctuations of the squeezing angle. High optical efficiency and accurate control of the squeezing phase are therefore essential.

4.4.3.6 Summary

LIGO does not evade the uncertainty principle. Instead, squeezed light places less quantum uncertainty in the quadrature that limits a particular part of the measurement and more uncertainty in the conjugate quadrature.

Frequency-dependent squeezing extends this idea by rotating the squeezing ellipse as a function of frequency. It reduces the relevant quantum noise at both low and high frequencies, allowing weaker gravitational-wave signals to be distinguished from the quantum fluctuations of the light.

This section summarizes LIGO's use of vacuum fluctuations entering the unused interferometer port, squeezed states to reduce shot noise, and frequency-dependent squeezing to control the tradeoff between high-frequency phase noise and low-frequency radiation-pressure noise.

LIGO reports that frequency-dependent squeezing has been operating since the observing run began in May 2023 and improves sensitivity across its gravitational-wave frequency range.

Bibliography

- [1] J. R. Rydberg, The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science **29**, 331 (1890), <https://doi.org/10.1080/14786449008619945> .
- [2] W. Gerlach and O. Stern, Zeitschrift für Physik **9**, 349 (1922).
- [3] C. Davisson and L. H. Germer, Nature **119**, 558 (1927).
- [4] R. A. Millikan, Phys. Rev. **7**, 355 (1916).
- [5] A. H. Compton, Phys. Rev. **21**, 483 (1923).